

Optimal Carbon Tax with Dynamic Skill Heterogeneity

Alexandre Pavlov*

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Abstract

How should a government design a carbon tax when it is not only concerned by mitigating carbon emissions, but also income redistribution? This article extends a dynamic Mirrlees taxation model, typically used to characterize optimal labor wedges and savings wedges, to study the optimal commodity wedge between two goods, one of which is polluting. First, starting with the basic case where there are two non-polluting goods, I derive the optimal commodity wedge with non-separable preferences in a dynamic environment. Second, in the case where one of the goods is polluting and pollution damages utility, I derive a modified optimal carbon tax whose output damages component remains Pigouvian and whose utility damages component is distorted by the labor wedge as well as the gap in complementarity in utility between each good and labor supply. Third, because real carbon taxes are anonymous linear prices rather than the history-dependent wedge the theory prescribes, I solve the economy when the commodity instrument must be uniform. The restriction returns the two-good problem to a one-good dynamic Mirrlees problem in expenditure: the optimal uniform tax is a substitution-weighted average of the history-dependent wedges, the labor and savings margins pick up a fiscal-interaction term, and the welfare cost of uniformity is the substitution-weighted dispersion of the unconstrained wedge, which vanishes under weak separability. With the externality the optimal uniform price splits additively into the social cost of carbon and a screening component, so uniformity constrains the redistributive part of the carbon price and never

*Université de Montréal: alexandre.pavlov@umontreal.ca

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the Pigouvian part.

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1 Introduction

Although an economist's favorite, carbon pricing remains shunned by the wider electorate on the charge that it unfairly burdens low-income households. France's *gilets jaunes* protests and Canada's 2025 repeal of its consumer carbon tax are the most recent examples of this resistance. This mismatch between expert and public opinion might in part be explained by the analytical framework governments rely on: integrated assessment models that they use to decide on environmental policy (Environment and Climate Change Canada (2023), Interagency Working Group on Social Cost of Greenhouse Gases (2021)) treat the world or each country as a single representative agent and are therefore not designed to weigh distributional concerns.

Whether carbon taxes are in fact regressive has been contested for three decades, and the contest turns on the life cycle. Ranked by annual income, taxes on gasoline and other energy fall far more heavily on the poor; ranked by lifetime or permanent income, the burden is much flatter, because households smooth consumption and today's low earners include the young and the temporarily unlucky (Poterba 1989; Hassett et al. 2009). The welfare-relevant burden of a carbon tax is a lifetime object, not an annual one. A model that speaks to the regressivity debate on its own terms must therefore be a lifecycle model.

This debate is entirely positive. It measures the incidence of existing carbon taxes and never asks what carbon tax a government *should* levy once it sees the whole life cycle. That normative question is the one I take up. What carbon tax is optimal over the life cycle, when the government redistributes through a non-linear income tax, earnings vary with age and private luck, and households save to smooth consumption? Should the carbon price lean against the lifetime incidence of energy, or does that task fall to the income tax?

No existing literature answers this, because each of the three relevant strands is missing an ingredient. The second-best environmental literature, from Sandmo (1975) through Barrage (2019) to the heterogeneous-agent treatments of Douenne et al. (2026) and Douenne et al. (2025), works with linear instruments and observable types, so it carries no non-linear income tax and no savings wedge. The static Mirrlees environmental literature (Cremer et al. 1998; Jacobs and de Mooij 2015) has the non-linear income tax but only one period, hence no life cycle, no savings, and no age profile of earnings. The dynamic Mirrlees literature (Goloso et al. 2003; Goloso et al. 2016) has private information, history-dependent lifecycle wedges, and the inverse Euler equation, but no externality and a single good. The age-dependent taxation literature showed that conditioning income taxes on age yields large welfare gains (Weinzierl 2011), yet never carried the logic to carbon. The lifecycle optimal carbon tax is the empty cell

these literatures circle without entering.

I answer the question in a dynamic Mirrlees economy with two goods, one clean and one dirty, in which agents draw privately observed skills over the life cycle and their consumption of the dirty good generates a stock externality. To my knowledge this is the first fully Mirrleesian derivation of the optimal carbon tax, with private information, non-linear history-dependent taxes, and a positive savings wedge. The central finding is a *separation*. Redistributing through the optimal non-linear income tax restores the split between corrective and redistributive taxation that the Ramsey second-best breaks: under weak separability, the carbon tax equals the social cost of carbon, the labor wedge keeps the Golosov et al. (2016) ABCD form, and the savings wedge keeps its inverse-Euler form. Three margins do three jobs without interacting. The carbon price should not be bent for lifetime distribution; the non-linear income tax and the climate dividend do that work. The below-Pigou prescription of the distortionary-taxation tradition is thus an artifact of the linear-instrument restriction, and it disappears once the income tax is optimized in a dynamic private-information economy.

I quantify the separation by solving the mechanism numerically, calibrated to US data. The optimal wedge on energy is the social cost of carbon times a history-dependent Corlett–Hague premium: it starts at the Pigouvian level, rises with age as the labor wedge does, and reaches about nine percent by the end of the working life. That premium is small in welfare. A flat carbon tax at the social cost of carbon captures ninety-eight percent of the welfare of the full non-linear optimum, and the entire Mirrleesian fine structure of the wedge, its age profile, history dependence, and progressivity in skill, is worth a few thousandths of a percent of consumption. This is the separation in welfare: pricing carbon at the social cost of carbon and redistributing through the income tax is close to optimal even when preferences are non-separable and energy is a leisure complement.

The deviations that do arise away from the Pigouvian benchmark are small, signed by complementarity gaps between each good and labor, and progressive in skill. Late in the working life the optimal energy wedge reaches about seventy percent for the top skill quintile against fifty-seven for the bottom, both above the Pigouvian level near fifty: the optimal carbon wedge rises with skill rather than falling. The regressivity of *real* carbon taxes is therefore a property of flat instruments and of energy’s status as a necessity, not of the optimum, and it is the normative counterpart the positive incidence debate never supplied. The results complement Douenne et al. (2026) and Douenne et al. (2025), who reach a near-Pigouvian conclusion from the Ramsey side; here I derive the preference conditions that make the separation exact and the channels that break it.

The paper proceeds as follows. Section 2 places the argument in the literature. Section 4 derives the optimal commodity wedge in a dynamic Mirrlees economy without an externality, isolating the single margin whose vanishing is the separation. Section 5 introduces the externality and states the separation and the modified social cost of carbon. Section 6 restricts the commodity

instrument to a uniform anonymous price, the form real carbon taxes take, and prices the welfare cost of the restriction. Section 7 solves the mechanism numerically, calibrated to US data, and measures the size of the optimal wedge and the welfare cost of simpler carbon-pricing rules, and section 8 concludes.

2 Literature review

This paper sits between two literatures that reach opposite verdicts on how distortionary taxation should bend the carbon price, and draws on a third that has the tools to adjudicate but has never been aimed at the environment. The disagreement is consequential. Should a government facing a distortionary tax system price carbon below marginal damages, because the environmental tax compounds distortions elsewhere, or does that markdown dissolve once the income tax is itself set optimally? The answer decides whether tight fiscal constraints call for timid or full carbon pricing, and it turns on how flexible the redistribution instrument is.

The Ramsey tradition answers that the markdown is real and grows with the distortion. Sandmo (1975) showed that with linear instruments the optimal pollution tax is additive, marginal damages entering the polluting good’s formula alone, but divided by the marginal cost of public funds, which exceeds one when labor is taxed distortionarily: the optimal pollution tax then lies strictly below marginal damages. Bovenberg and de Mooij (1994) traced the gap to a tax-interaction effect, by which the environmental tax amplifies the pre-existing labor-supply distortion, and Parry et al. (1999) placed the optimal rate at 63 to 78 percent of marginal damage. Carried into the dynamic climate-economy models that discipline policy, the verdict survives. Golosov et al. (2014) supply the closed-form representative-agent benchmark, and Barrage (2019) puts the optimal dynamic carbon tax 8 to 24 percent below it, driven by the same public-funds division and by rigid capital-tax instruments. This whole arc, though, is representative-agent or observable-type: inequality enters neither the objective nor the incentive constraints.

Adding heterogeneous agents to this dynamic Ramsey frontier softens the verdict, and these are the closest antecedents to what follows. Douenne et al. (2026) solves a heterogeneous-agent Ramsey problem with a calibrated climate module and linear instruments, and finds the optimal carbon tax essentially Pigouvian, inequality lowering it by only about four percent; Douenne et al. (2025) shows the result robust to uninsurable income risk and tight fiscal constraints. The distributional stakes they engage are real: Känzig (2023) finds carbon-pricing shocks fall roughly three times harder on low-income than on high-income households, and the dispersion of energy spending *within* income groups exceeds the dispersion *across* them (Cronin et al. 2019; Pizer and Sexton 2019), precisely the heterogeneity a type-specific wedge would respond to. Restricting the instrument menu can even reverse the near-Pigouvian verdict: van der Ploeg et al. (2025) show that a uniform carbon tax paired only with a uniform dividend can optimally exceed marginal damages, and Hänsel et al. (2022) that a mandated uniform carbon tax makes the

optimal income tax more progressive. A parallel policy debate, meanwhile, presses for carbon prices that differ across individuals and over their cumulative choices (Chancel 2022; Dietsch 2024), a demand that no incentive-compatible optimal-tax analysis has yet grounded. All of this, though, remains within the Ramsey paradigm, with linear instruments, observable types, and period-by-period planning, so it carries neither a nonlinear income tax nor an inverse-Euler savings wedge.

A parallel static Mirrleesian literature argues the opposite: that the Ramsey markdown is not a robust feature of distortionary taxation but an artifact of the linear-instrument restriction. Cremer et al. (1998) showed that when individual consumption is observable and the nonlinear income tax is optimized, Sandmo additivity survives and the public-funds division disappears, and Kaplow (2012) made the argument cleanest: any environmental reform can be rendered distribution-neutral by adjusting the income tax type by type, so that a move to the Pigouvian rate is Pareto-improving. Jacobs and de Mooij (2015) formalized it, showing that with an optimized nonlinear income tax the marginal cost of public funds equals one and the optimal pollution tax equals marginal damages, provided preferences are weakly separable in the externality. When separability fails the tax deviates in signable directions: above Pigouvian for a leisure complement, the Corlett–Hague case substantiated for gasoline by West and Williams III (2007), below for a necessity with non-homothetic Engel curves (Jacobs and van der Ploeg 2019), and either way once the redistributive role of commodity taxation is in play (Christiansen 1984). The calibrated two-good static-Mirrlees treatment nearest to the environment used here is Cremer et al. (2010). All of this is static, however: it has the nonlinear income tax but no life cycle, no savings, and no age profile of earnings, so it cannot discipline the time path of the carbon tax that integrated assessment models require.

The tools for a dynamic treatment come from a third literature, which built the private-information theory of optimal taxation but abstracted from externalities. Golosov et al. (2003) established the inverse Euler equation, under which private saving relaxes the high type’s mimicking incentive so that the optimal savings wedge is strictly positive, breaking the Chamley–Judd zero-capital-tax result. Kapička (2013) extended the first-order approach to persistent shocks, Farhi and Werning (2013) characterized the labor wedge over the life cycle, and Golosov et al. (2016) gave its tightest form, the ABCD decomposition this paper builds on, with Atkinson–Stiglitz uniform commodity taxation surviving within each period under weak separability. That the same apparatus has already been turned on lifetime-varying taxes without reaching the environment makes the gap conspicuous: Weinzierl (2011) finds large welfare gains from conditioning the income tax on age alone. The nearest neighbor with an externality, Chen and Ren (2024), embeds a generic consumption externality in a two-period economy but has neither a genuine cross-agent pollutant with output and utility damages nor an explicit clean/dirty commodity wedge, and the closest environmental hybrid, Crampes and Ladoux (2024), prices energy efficiency in a two-period Mirrleesian setting without persistent skill shocks or the inverse-Euler structure.

These three literatures circle a common empty cell. The Ramsey arc has reached a dynamic heterogeneous-agent frontier but stays bound to linear instruments and observable types; the static Mirrleesian literature has the analytical Pigouvian-irrelevance result but only one period; the dynamic Mirrleesian apparatus has private information, history-dependent wedges, and the inverse Euler equation but no externality and a single good. This paper fills the cell, and its three contributions map onto the three strands. Section 4 carries the dynamic Mirrleesian apparatus from one consumption good to two. Section 5 extends the Jacobs and de Mooij (2015) irrelevance theorem to a dynamic private-information economy and, under non-separability, signs the deviations Ramsey treatments cannot recover. Section 6 returns the carbon price to the anonymous linear form real policy uses and prices the cost of that restriction. The result complements Douenne et al. (2026) rather than competing with it: where they measure, quantitatively and from the Ramsey side, that the optimal carbon tax is near-Pigouvian, this paper derives, analytically and from the Mirrleesian side, the preference conditions under which corrective and redistributive taxation separate exactly, and the channels that break the separation.

3 Environment

The economy is a two-good extension of the dynamic Mirrlees setting of Golosov et al. (2003), Farhi and Werning (2013), and Golosov et al. (2016). The two goods are a clean numéraire and a polluting good. To isolate how household income inequality shapes the optimal carbon tax, I abstract from international and intergenerational equity. Hence, I only consider a single country and a single generation.

The economy lasts $T + 1$ periods. Each household consumes a clean good c_t and a polluting good q_t , supplies labor l_t , and is exposed to the aggregate pollution stock S_t .

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right]$$

$\beta \in (0, 1)$ is the discount factor and the clean good c is the numéraire, whose price is normalized to 1. To simplify notation, I assume the price of the dirty good is also 1. Utility is time-separable across periods but may be non-separable across its arguments. Consuming the dirty good generates emissions that accumulate into a persistent pollutant stock S_t , which evolves with current emissions and past stocks,

$$S_t = S(S_0, S_1, \dots, S_{t-1}, Q_t),$$

where $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] dF_0(\theta)$ is aggregate consumption of the dirty good at date t . This characterization allows current emissions to affect future pollutant accumulation, for example when last period's stock remains in the atmosphere after linear depreciation δ ,

$$S_t = \delta S_{t-1} + Q_t.$$

I leave the pollutant accumulation function otherwise unrestricted. To keep the problem recursive, I assume emissions stop affecting the stock after s periods,

$$S_t = S(S_{t-s}, \dots, S_{t-1}, Q_t).$$

Aside from the presence of a polluting good, the model follows the basic setup of the dynamic Mirrlees literature. At $t = 0$, agents draw an initial skill (type) θ_0 from a distribution $F_0(\theta)$. Later skills follow a Markov process $F_t(\theta|\theta_{t-1})$. At period t , each agent knows their skill draws up to time t , which I denote by a skill history vector $\theta^t = (\theta_0, \dots, \theta_t)$. Θ^t is the set of all possible skill histories.

An agent of type θ_t who supplies l_t units of labor produces $y_t = \theta_t l_t$ units of output. Everyone observes total output, but skill θ_t and labor supply l_t are private. Consumption of both goods is also publicly observable¹. Let $c_t(\theta^t), q_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ and $y_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ be the agent's allocation of the clean good, the polluting good, and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \rightarrow \Theta^t$ describe how an agent of type θ_t chooses to report their type, and let Σ^t be the set of all possible reporting strategies. Agents can use capital to transfer resources between periods, earning gross return $R > 0$. The social welfare function is

$$\int_0^\infty \alpha(\theta) \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right] dF_0(\theta)$$

$\alpha(\theta)$ is a non-negative Pareto weight on the welfare of an agent with initial skill θ . If the planner is utilitarian, $\alpha(\theta) = 1$ for all θ . Let U_c and U_q denote the partial derivatives with respect to the clean and polluting goods, and U_l with respect to labor. Since $y = \theta l$, I can write $U(c, q, l, S)$ as $U(c, q, \frac{y}{\theta}, S)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ .

Assumption 3.0.1 (preferences).

U is twice continuously differentiable with $U_1, U_2 > 0$, $U_l, U_S < 0$, $U_{11} \leq 0$, $U_{22} \leq 0$, $U_{ll} \leq 0$, jointly concave in (c_1, c_2, l) , and satisfies Inada conditions in (c_1, c_2) . Single crossing holds:

$$\frac{\partial}{\partial \theta} \frac{U_l(c_1, c_2, y/\theta)}{U_1(c_1, c_2, y/\theta)} \geq 0 \quad \text{for all } (c_1, c_2, y, \theta).$$

Assumption 3.0.2 (shock process).

For all $t < \hat{T}$, f_t is differentiable in both arguments with $f'_t = \partial f_t / \partial \theta$ and $f_{2,t} = \partial f_t / \partial \theta_{t-1}$. For all θ_{t-1} ,

$$\varphi_t(\theta | \theta_{t-1}) \equiv \frac{\theta_{t-1} \int_\theta^\infty f_{2,t}(x | \theta_{t-1}) dx}{\theta f_t(\theta | \theta_{t-1})}$$

is bounded, and $\lim_{\theta \rightarrow \infty} \frac{1 - F_t(\theta | \theta_{t-1})}{\theta f_t(\theta | \theta_{t-1})}$ is finite.

Any admissible allocation must be incentive-compatible: the agent must weakly prefer truth-telling to any other strategy. For all strategies and skill histories,

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right]$$

¹This is a plausible assumption for some goods such as electricity and air travel. For a wider set of goods whose consumption cannot be as easily observed, the planner's optimum will be further constrained and the results in this paper no longer hold.

I follow the first-order approach of Kapička (2013), Farhi and Werning (2013), and Golosov et al. (2016), replacing the global incentive constraints with their local counterpart. Let $u_t(\theta^t)$ be the lifetime utility an agent obtains from reporting truthfully at every date. Truth-telling is locally optimal when the envelope condition holds.

$$\dot{u}_t(\theta^t) = U_\theta \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) + \beta w_{2,t}(\theta^t)$$

$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta_{t+1}) \partial_{\theta_t} f_{t+1}(\theta_{t+1} | \theta_t) d\theta_{t+1}$ is the marginal continuation value of type θ^t . Together with a monotonicity requirement on the assigned allocation, and under Assumption 3.0.1, this local condition is necessary for incentive compatibility.

The aggregate resource constraint must hold in every period,

$$\int_0^\infty \mathbb{E}_0[c_t(\theta^t) + q_t(\theta^t) | \theta] dF_0(\theta) + K_{t+1} \leq \int_0^\infty \mathbb{E}_0[y_t(\theta^t) | \theta] dF_0(\theta) + RK_t,$$

where K_t is the aggregate capital stock, with K_0 given and $K_{T+1} = 0$. Dividing each period's constraint by R^t , summing over t , and using these boundary conditions to telescope the capital terms consolidates the sequence into a single intertemporal constraint,

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{c_t(\theta^t) + q_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta).$$

Combining the planner's objective with the incentive-compatibility and resource-feasibility constraints yields the following planner's problem,

$$\begin{aligned} & \max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t, t=0, \dots, T}} \int_0^\infty \alpha(\theta) \left(\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \right) dF_0(\theta) \\ & \text{s.t.} \\ & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \\ & \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{c_t(\theta^t) + q_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) \\ & S_t = S(S_{t-s}, \dots, S_{t-1}, Q_t) \end{aligned}$$

Three wedges describe the optimal allocation: the labor wedge, the savings wedge, and the commodity wedge.

$$\begin{aligned}
1 - \tau_t^y(\theta^t) &\equiv -\frac{U_l\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\theta_t U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)} \\
1 - \tau_t^s(\theta^t) &\equiv \frac{1}{\beta R} \frac{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\mathbb{E}_t\left[U_c\left(c_{t+1}(\theta^{t+1}), q_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}, S_{t+1}\right)\right]} \\
1 + \tau_t^q(\theta^t) &\equiv \frac{U_q\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}
\end{aligned}$$

I also define the following five elasticities:

- $\epsilon_t(\theta) \equiv \frac{U_{ll,t}(\theta) l_t(\theta)}{U_{l,t}(\theta)}$, the inverse Frisch elasticity of labor supply;
- $\sigma_t(\theta) \equiv -\frac{U_{cc,t}(\theta) c_t(\theta)}{U_{c,t}(\theta)}$, the inverse elasticity of intertemporal substitution of the numéraire;
- $\sigma_{cq,t}(\theta) \equiv -\frac{U_{cq,t}(\theta) q_t(\theta)}{U_{c,t}(\theta)}$, the cross-curvature of the numéraire's marginal utility;
- $\gamma_{c,t}(\theta) \equiv \frac{U_{cl,t}(\theta) l_t(\theta)}{U_{c,t}(\theta)}$, the complementarity of the clean good with labor supply;
- $\gamma_{q,t}(\theta) \equiv \frac{U_{ql,t}(\theta) l_t(\theta)}{U_{q,t}(\theta)}$, the complementarity of the polluting good with labor supply.

4 Optimal dynamic multi-good wedges

Before adding pollution, I characterize the commodity wedge in a clean two-good economy. Section 5 shows that the externality leaves this structure intact, only adding a corrective term. The problem can be written recursively, using the results in Kapička (2013)².

$$\begin{aligned}
V_t(\hat{w}, \hat{w}_2, \theta_-) &= \min_{c, q, y, u, w, w_2} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta) \right] f_t(\theta | \theta_-) d\theta \\
&\quad \text{s.t.} \\
\dot{u}(\theta) &= U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta} \right) + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta) \bar{\alpha}_t(\theta) f_t(\theta | \theta_-) d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta | \theta_-) d\theta \\
u(\theta) &= U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta} \right) + \beta w(\theta)
\end{aligned}$$

Here $\bar{\alpha}_t(\theta)$ is the continuation Pareto weight attached to delivering utility to type θ . Solving the problem yields the three wedges below: the labor and savings wedges take their one-good form in the marginal utility of the numéraire, and a commodity wedge between the numéraire and the polluting good is the structural novelty.

Result 4.0.1 (Optimal wedges).

The labor wedge is

$$\frac{\tau^y}{1 - \tau^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

$$A_t(\theta) = 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_\theta^\infty \exp \left[\int_\theta^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_\theta^\infty \exp \left[- \int_\theta^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}$$

²Kapicka considers that consumption comes from an arbitrary convex and compact set, which means it is not restricted to a scalar and can very well be a vector of multiple consumption goods.

Define $\widehat{U}_{c,t} \equiv U_{c,t}/(1-\chi_t)$, with $\chi_t \equiv \frac{\gamma_{c,t}}{1+\epsilon_t-\gamma_{c,t}} \frac{\tau_t^y}{1-\tau_t^y}$ as the complementarity-weighted labor wedge. The savings wedge is

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right],$$

The commodity wedge is

$$\frac{\tau^q}{1+\tau^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1+\epsilon_t - \gamma_{c,t}} \frac{\tau^y}{1-\tau^y},$$

Proof. See Appendices A.1, A.2, and A.3. □

The labor wedge is Golosov et al. (2016)'s decomposition term for term, now written in the marginal utility of the numéraire. Three of the four pieces are unchanged: $A_t = 1 + \epsilon_t - \gamma_{c,t}$ scales the wedge by the responsiveness of labor supply, B_t is the hazard ratio that weighs the benefit of relaxing the incentive constraint at θ , which accrues to the mass $1 - F_t(\theta)$ of higher types, against its cost at the local mass $\theta f_t(\theta)$, and D_t carries the backward-looking persistence inherited from past distortions. Only C_t , the social value of the resources extracted from higher types, changes. Its exponent measures how the numéraire's marginal utility evolves as the planner takes resources up the type distribution, and with two goods that evolution runs through both margins: the cross-curvature $\sigma_{cq,t} \dot{q}_t/q_t$ now sits beside the own-curvature $\sigma_t \dot{c}_t/c_t$. The reason is immediate: taking resources from a higher type cuts its consumption of both goods and, when $U_{cq} \neq 0$, the fall in the dirty good moves the marginal value of the numéraire in which the wedge is denominated. Under separability between the goods the cross term vanishes and C_t returns to its one-good form. The second good thus leaves the structure of the labor wedge intact, merely reweighing the income effect inside C_t .

The savings wedge is once more an inverse Euler equation, now in the incentive-adjusted marginal utility $\widehat{U}_{c,t} = U_{c,t}/(1-\chi_t)$. Delivering a util to a type through the numéraire costs $1/U_{c,t}$ in resources, but a marginal unit of the numéraire also shifts the effort a local mimicker would supply, at incentive cost $\chi_t/U_{c,t}$; the planner equalizes the present value of this adjusted cost across dates and states, just as the classical inverse Euler equation equalizes $1/U_c$ under separable preferences. The adjustment χ_t is the complementarity-weighted labor wedge, which grows with the concurrent labor distortion and disappears when the numéraire is separable from labor. This is a counterpart to Hellwig (2021)'s generalized inverse Euler condition, which accommodates non-separability through a change in probability measure over types. His tilt of the measure and my factor $1-\chi_t$ on marginal utility are essentially the same, and both collapse

to the textbook rule once consumption and labor separate. The adjustment is a property of the numéraire's complementarity with labor however and not of the second good.

The commodity wedge is the labor wedge weighed by the ratio of the complementarity gap and the labor response. Its sign follows the Corlett-Hague logic: the planner taxes the good that is the weaker complement of labor. When the two goods are equally labor-complementary the gap closes and the wedge is zero, the dynamic Atkinson-Stiglitz benchmark. What is new relative to the static Mirrleesian commodity-tax problem is that the wedge is a dynamic object. Hence, optimal commodity taxation ought to be history-dependent if preferences are non-separable.

5 Optimal dynamic non-linear wedges with externalities

I now introduce the externality. The global incentive constraints in the planner's problem can be substituted using the results in Kapička (2013) to yield the following recursive problem.

$$\begin{aligned}
V_t(\hat{w}, \hat{w}_2, \theta_-, S_-) = & \min_{c, q, y, u, w, w_2, S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, S) \right] f_t(\theta|\theta_-) d\theta \\
& \text{s.t.} \\
& \dot{u}(\theta, S) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w_2(\theta) \\
& \hat{w} = \int_0^\infty u(\theta) \bar{\alpha}_t(\theta) f_t(\theta|\theta_-) d\theta \\
& \hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta \\
& u(\theta, S) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w(\theta) \\
& S = S \left(\dots, \int q(\theta) f(\theta) d\theta \right)
\end{aligned}$$

Solving this problem, the structure of income taxes is once again unchanged. However, the commodity wedge now features what amounts to a modified Pigouvian term where marginal damages of pollution to utility are adjusted for the planner's fiscal concerns.

Result 5.0.1 (Optimal corrective wedge).

The optimal corrective wedge between the "clean" good and the "polluting" good is

$$\frac{\tau_t^q - SCC_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right]$$

where

$$SCC_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + \left(\frac{\tau_{t+j}^y}{1 - \tau_{t+j}^y} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

with $\Phi_j \equiv \partial S_{t+j} / \partial Q_t$ the effect of current aggregate emissions on the date- $(t+j)$ pollution stock ($\Phi_j = 0$ for $j > s$ by assumption) and the expectation taken over future skill histories. The bracket is the no-pollution commodity wedge of Result 4.01 term for term, with A_t, B_t, C_t, D_t the labor-wedge terms. The externality changes only the left-hand side, replacing τ_t^q with the deviation $\tau_t^q - SCC_t$.

Proof. See appendix B.1. □

The optimal corrective wedge in (5.0.1) is common to all types and decomposes into two economically distinct components. We have a familiar Mirrleesian component, which takes the same shape as the no-pollution case, and a new corrective component which features a modified Pigouvian term SCC_t . However, the commodity wedge now operates on the deviation $\tau_t^q - SCC_t$ rather than on τ_t^q itself. The SCC_t is the discounted sum of marginal utility damages of pollution multiplied by a fiscal adjustment factor. Whenever the labor wedge is absent or the pollution stock interacts with labor identically to the clean good, the adjustment factor equals one and the SCC_t reduces to the textbook Pigouvian social cost of carbon. If $\xi_j > \gamma_{c,j}$, abatement provides a fiscal bonus, partially undoing the labor wedge, and the SCC exceeds the pure Pigouvian level. If $\xi_j < \gamma_{c,j}$, abatement worsens the labor distortion and the SCC is under the Pigouvian level. This is the "virtual subsidy" channel of Jacobs and de Mooij (2015) generalised here to a dynamic setting.

When the Mirrleesian motive disappears, either because goods are equally complementary with labor or because preferences are separable between consumption and labor, the planner sets $\tau_t^q = SCC_t$ exactly. That is, unlike in a dynamic Ramsey environment such as Barrage (2019) or Douenne et al. (2026), the presence of positive labor and capital taxation does not imply a deviation from the standard Pigouvian rule unless there is non-separability between labor supply and the other variables.

6 Uniform commodity wedges

The commodity wedge of Sections 4 and 5 is history-dependent: it tracks the labor wedge node by node. Real commodity and carbon taxes are not. An excise or a carbon price is an anonymous linear price, a single figure per period that every buyer faces regardless of earnings history. This section re-solves the economy under exactly that restriction. The planner keeps the fully non-linear, history-dependent income and savings instruments, but the consumer price of the dirty good must be uniform within each period. I ask how all three margins are set and what the restriction costs.

The restriction does more than make the problem tractable: it returns the problem to the one-good case. When the agent splits assigned expenditure privately at the common price, the planner effectively assigns within-period expenditure and output, and the agent's period payoff is the indirect utility over expenditure. The type still enters only through $l = y/\theta$, so the envelope condition that drives the whole apparatus is untouched, and every result of Section 4 applies with composite statistics. The two-good structure survives in exactly two places. The composite statistics are marginal-budget-share aggregates of the primitive ones, and the resource cost now carries commodity-tax revenue, whose dependence on the assignment adds a fiscal-interaction term to the labor and savings conditions. This reduction is also the object the quantitative section solves: the linear-commodity solver of Section 7 is this reduction made numerical.

6.1 The restricted problem and its reduction

The planner sets a uniform per-unit tax T_t on the dirty good. Its producer price is one, so its consumer price is $1 + T_t$, common to all agents at date t . After each history the planner assigns within-period expenditure $x_t(\theta^t) \geq 0$, measured at consumer prices, and output $y_t(\theta^t)$; the agent splits x between the two goods privately,

$$\tilde{U}(x, l; T) \equiv \max_{c, q \geq 0} U(c, q, l) \quad \text{s.t.} \quad c + (1 + T)q \leq x,$$

with Marshallian demands $c^d(x, l, T)$ and $q^d(x, l, T)$ pinned down by the tangency $U_q = (1+T)U_c$ and the budget $c + (1 + T)q = x$. The realized commodity wedge is $1 + \tau_t^q = U_q/U_c = 1 + T_t$, common to every agent: the uniform wedge is the uniform tax.

Lemma 6.1.1 (envelope properties and reduction).

At an interior split, $\tilde{U}_x = U_c$, $\tilde{U}_l = U_l$, and $\tilde{U}_T = -q^d U_c$ (Roy's identity). An assignment $\{x_t(\theta^t), y_t(\theta^t)\}$ with common prices $\{1 + T_t\}$ is incentive compatible if and only if it is incentive compatible in the one-good economy of Section 4 with period utility $\tilde{U}(x, y/\theta; T_t)$, and the

envelope condition is

$$\dot{u}(\theta) = -\frac{\tilde{U}_l l}{\theta} + \beta w_2(\theta),$$

exactly as before with $\tilde{U}_l$ in place of U_l .

Proof. See appendix C.1. □

Lemma 6.1.1 answers whether the framework accommodates a uniform wedge. The restriction does not strain the first-order approach; it returns the problem to the one-good case for which the approach was built. A deviator who misreports receives an expenditure-output pair and splits it optimally at the same common price, so the deviation space is that of a one-good economy in $\tilde{U}$. Because $\tilde{U}_x = U_c$ and $\tilde{U}_l = U_l$ at the chosen bundle, the labor and savings wedges τ_t^y and τ_t^s are numerically the same objects as in Section 3, directly comparable across the constrained and unconstrained solutions. All that changes is which statistics of $\tilde{U}$ inherit which structure, and the appearance of commodity revenue in the resource cost.

6.2 Composite statistics

Every formula below is stated in composite statistics of $\tilde{U}$, defined as the Section 3 statistics were but on expenditure rather than on the numéraire,

$$\tilde{\epsilon} \equiv \frac{\tilde{U}_{ll} l}{\tilde{U}_l}, \quad \tilde{\gamma} \equiv \frac{\tilde{U}_{xl} l}{\tilde{U}_x}, \quad \tilde{\sigma} \equiv -\frac{x \tilde{U}_{xx}}{\tilde{U}_x}, \quad \tilde{A} \equiv 1 + \tilde{\epsilon} - \tilde{\gamma}.$$

Let $m_q \equiv (1+T) \partial_x q^d$ be the marginal budget share of the dirty good, the fraction of a marginal dollar of expenditure spent on it, and let $s_{qq} \equiv \partial q^d / \partial (1+T)|_u < 0$ be its compensated own-price response.

Lemma 6.2.1 (composite statistics).

With $\mathcal{D} \equiv U_{qq} - 2(1+T)U_{cq} + (1+T)^2 U_{cc} < 0$,

$$\tilde{\gamma} = (1 - m_q) \gamma_c + m_q \gamma_q, \quad \tilde{\epsilon} = \epsilon - \frac{(1+T)^2 U_c^2 (\gamma_c - \gamma_q)^2}{l |U_l| |\mathcal{D}|} \leq \epsilon.$$

Under weak separability, $\gamma_c = \gamma_q$, so $\tilde{\gamma} = \gamma_c = \gamma_q$, $\tilde{\epsilon} = \epsilon$, and the reduced problem is that of Section 4 exactly.

Proof. See appendix C.2. □

Both expressions are intuitive. The composite complementarity is the marginal-spending-weighted average of the two goods' complementarities: a marginal dollar of expenditure is split $(1 - m_q, m_q)$ across the goods, so the marginal utility it buys co-moves with labor accordingly. The inequality on $\tilde{\epsilon}$ is a Le Chatelier property. The agent's freedom to re-optimize the split cushions labor shocks, so the effective disutility of labor is less convex, and delivering expenditure is a weakly better hedge than delivering a rigid bundle. Both corrections are proportional to the same complementarity gap $\gamma_c - \gamma_q$ that drives the unconstrained wedge of Section 4: everything the restriction does vanishes when the two goods are equally complementary with labor.

6.3 The three margins

Let $\tilde{\Lambda}_t \equiv \psi \tilde{U}_x / (\theta f)$ be the normalized shadow price of the local incentive constraint in the reduced problem, the counterpart of the object behind the Section 4 wedges (here ψ is the costate on the envelope condition, as in appendix A.1).

Result 6.3.1 (labor wedge under a uniform commodity tax).

At every history,

$$\frac{\tau_t^y}{1 - \tau_t^y} + \frac{T_t}{1 - \tau_t^y} \frac{dq_t^d}{dy} \Big|_u = \tilde{A}_t \tilde{\Lambda}_t, \quad \frac{dq_t^d}{dy} \Big|_u \equiv (1 - \tau^y) \partial_x q^d + \frac{\partial_l q^d}{\theta},$$

where $dq^d/dy|_u$ is the utility-compensated response of dirty-good demand to earnings: the change in q^d when output rises by one unit and expenditure rises by $(1 - \tau^y)$ units so as to hold $\tilde{U}$ fixed.

Proof. See appendix C.4. □

The reduced problem prices a comprehensive wedge, not the income-tax wedge alone. When an agent earns a marginal unit of output on a compensated path, the planner collects τ_t^y through the income margin and $T_t dq^d/dy|_u$ through the commodity margin; their sum $\Omega_t \equiv \tau_t^y / (1 - \tau_t^y) + T_t / (1 - \tau_t^y) \cdot dq^d/dy|_u$ is what the Golosov–Troshkin–Tsyvinski-form condition prices. The sign of the correction follows Christiansen. If the dirty good is a leisure complement, $\gamma_q < \gamma_c$, then $\partial_l q^d < 0$ and $dq^d/dy|_u < 0$: working more, compensated, shifts spending away from the leisure-complementary good. With a positive uniform tax the measured income-tax wedge then exceeds the naive $\tilde{A}_t \tilde{\Lambda}_t$. The economics is that discouraging labor is fiscally cheaper: the agent who works less spends more on the taxed good, and part of the forgone output returns as commodity revenue, so the planner tolerates a larger labor distortion for the same screening benefit.

Result 6.3.2 (optimal uniform commodity tax).

At every date,

$$T_t \mathbb{E}_0[s_{qq,t}] = \mathbb{E}_0[\tilde{\Lambda}_t l_t \partial_l q_t^d] \iff \frac{\tau_t^q}{1 + \tau_t^q} = \frac{\mathbb{E}_0[|s_{qq,t}| (\gamma_{c,t} - \gamma_{q,t}) \tilde{\Lambda}_t]}{\mathbb{E}_0[|s_{qq,t}|]},$$

where $\mathbb{E}_0$ is the unconditional population expectation over date- t histories θ^t . The uniform ad-valorem wedge is the compensated-substitution-weighted average of the unconstrained node-specific wedge $(\gamma_c - \gamma_q)\tilde{\Lambda}$ of Section 4.

Proof. See appendix C.5. □

The single instrument averages the schedule the planner would set node by node if it could, weighting each history by the compensated substitutability $|s_{qq}|$ at that node. Nodes where the consumer can easily substitute away from the dirty good carry more weight, exactly as in a many-person Ramsey rule. Two limits are immediate.

Corollary 6.3.1 (when uniformity is costless).

If preferences are weakly separable, $\gamma_c = \gamma_q$ pointwise, then $T_t = 0$ for every t , the fiscal terms vanish from every condition, and the restricted optimum coincides with the unconstrained optimum. This is the dynamic Atkinson–Stiglitz benchmark. More generally the restriction is slack at date t if and only if the unconstrained wedge $(\gamma_c - \gamma_q)\tilde{\Lambda}$ is constant across the date- t cross-section.

Corollary 6.3.2 (retirement).

For $t \geq \hat{T}$ there is no report and $\psi \equiv 0$, hence $\tilde{\Lambda} \equiv 0$ and $T_t = 0$. The screening component of the uniform tax dies with the labor margin; with the externality below, the dirty-good price converges to pure Pigouvian pricing for retirees.

Result 6.3.3 (savings wedge under a uniform commodity tax).

Define the incentive-plus-fiscal adjustment and the adjusted marginal utility of expenditure

$$\chi_t^F \equiv \underbrace{\tilde{\gamma}_t \tilde{\Lambda}_t}_{\text{incentive}} + \underbrace{\varpi_t m_{q,t}}_{\text{fiscal}}, \quad \varpi_t \equiv \frac{T_t}{1 + T_t}, \quad \hat{U}_{x,t} \equiv \frac{\tilde{U}_{x,t}}{1 - \chi_t^F}.$$

The generalized inverse Euler equation holds in $\hat{U}_x$,

$$\frac{1}{\hat{U}_{x,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\hat{U}_{x,t+1}} \right],$$

and to second order $\tau_t^s \approx \text{Var}_t(\ln \tilde{U}_{x,t+1}) + (\chi_t^F - \mathbb{E}_t \chi_{t+1}^F) + \text{Cov}_t(\chi_{t+1}^F, \ln \tilde{U}_{x,t+1})$.

Proof. See appendix C.6. □

The savings adjustment now has two parts. The incentive part $\tilde{\gamma}_t \tilde{\Lambda}_t$ is the Section 4 adjustment on the composite margin, present whenever expenditure is complementary with labor. The fiscal part $\varpi_t m_{q,t}$ is new. Delivering a marginal unit of expenditure returns the fraction $\varpi_t m_{q,t}$

to the planner as commodity revenue, so its net resource cost is $1 - \varpi_t m_{q,t}$, and the planner equalizes the present value of this net, incentive-adjusted cost across dates and states. Through the second-order expression the savings wedge picks up the term $\varpi_t m_{q,t} - \mathbb{E}_t[\varpi_{t+1} m_{q,t+1}]$, the expected change in the commodity-revenue content of spending. With a positive tax on a leisure complement, agents spend a larger share on the taxed good precisely when leisure is high, late in life and in retirement, so this term tends to be negative: deferring expenditure to high-leisure periods generates future commodity revenue, and the planner credits saving for it.

The labor wedge inherits the full dynamic structure of Section 4, now carried by the comprehensive wedge.

Result 6.3.4 (comprehensive-wedge decomposition).

At every history,

$$\Omega_t = \tilde{A}_t(\theta) \tilde{B}_t(\theta) \tilde{C}_t(\theta) + \beta R \Omega_{t-1} \tilde{D}_t(\theta),$$

where $\tilde{A}_t, \tilde{B}_t, \tilde{C}_t, \tilde{D}_t$ are the labor-wedge terms of Section 4 in composite statistics, with the redistributive weight in $\tilde{C}_t$ carrying the revenue term $1 - T_t \partial_x q^d - \lambda_{1,t} \bar{\alpha}_t \tilde{U}_x$ and a single composite consumption object in place of the cross-good exponent.

Proof. See appendix C.7. □

All of the qualitative structure survives the restriction, the hazard-driven shape through $\tilde{B}_t$, the persistence channel through $\tilde{D}_t$, but it governs the comprehensive margin Ω_t . Result 6.3.1 then dictates, given T_t , how Ω_t splits between the income-tax wedge and the commodity-revenue term.

6.4 The welfare cost of uniformity

Result 6.4.1 (cost of uniformity).

Let $\varpi_t^(\theta^t) \equiv (\gamma_{c,t} - \gamma_{q,t}) \tilde{\Lambda}_t$ denote the unconstrained node-specific ad-valorem wedge, and let $\text{Var}_t^{[s]}$ denote the $|s_{qq,t}|$ -weighted cross-sectional variance over date- t histories. To second order in the dispersion of ϖ_t^* , the date- t resource-equivalent loss from imposing a common tax rather than the node-specific schedule is*

$$\text{Loss}_t \approx \frac{1}{2} (1 + T_t)^2 \mathbb{E}_0[|s_{qq,t}|] \text{Var}_t^{[s]}(\varpi_t^*),$$

minimized at the uniform rate of Result 6.3.2.

Proof. See appendix C.8. □

The cost of not personalizing the commodity tax is the substitution-weighted cross-sectional variance of the unconstrained wedge. It vanishes under weak separability, where $\varpi^* \equiv 0$,

and it is small whenever the complementarity gap is small or the wedge $\varpi^* = (\gamma_c - \gamma_q)\tilde{\Lambda}$ is flat across the cross-section. This is the proportionality result of Section 4 priced: because the unconstrained wedge tracks the labor wedge, its dispersion is first-order in the dispersion of τ^y , and what that tracking is worth is exactly what uniformity forgoes. Every object in the formula is a demand-side or tax-side statistic, so it is a sufficient-statistics answer to the cost-of-mispricing question, and the near-sufficiency of a flat carbon price that the quantitative section measures is this variance being small.

6.5 The externality: uniformity binds Corlett–Hague, never Pigou

Reintroduce the externality of Section 5. Because the stock is aggregate and deterministic, it is invariant to reports and acts as a dated preference shifter, and pricing emissions converts the producer price of the dirty good to the emissions-inclusive $1 + \text{SCC}_t$. The uniform per-unit tax is now the screening component $T_t \equiv (1 + \tau_t^q) - (1 + \text{SCC}_t) = \tau_t^q - \text{SCC}_t$, the revenue wedge net of the part that pays for marginal damages.

Result 6.5.1 (additive split of the uniform price).

At an interior optimum of the restricted economy with the externality, all of the preceding results hold with $T_t = \tau_t^q - \text{SCC}_t$ and every statistic evaluated at the externality allocation. The optimal uniform consumer price splits additively,

$$1 + \tau_t^q = 1 + \text{SCC}_t + T_t^{\text{screen}}, \quad \frac{T_t^{\text{screen}}}{1 + \tau_t^q} = \frac{\mathbb{E}_0 \left[|s_{qq,t}| (\gamma_{c,t} - \gamma_{q,t}) \tilde{\Lambda}_t \right]}{\mathbb{E}_0 [|s_{qq,t}|]},$$

and SCC_t retains its Section 5 form with composite statistics.

Proof. See appendix C.9. □

The uniformity restriction never binds the Pigouvian component. The corrective part of the optimal dirty-good price was uniform already at the unconstrained optimum, since a ton of emissions carries the same social cost whoever emits it and the emissions margin cannot screen. What uniformity constrains is only the Corlett–Hague screening part, forced from a history-dependent schedule down to the substitution-weighted scalar of Result 6.3.2. For carbon policy the reading is sharp: a legislated uniform carbon price sacrifices nothing on the corrective margin, whatever is lost is lost on screening, and Result 6.4.1 prices that loss. This is the separation of Section 5 restated for the instrument governments actually use.

One caveat is proper to a restricted-instrument setting. The full-optimum production-efficiency logic of Diamond and Mirrlees (1971) no longer applies automatically once an instrument is constrained, and Naito (1999) shows that a constrained planner can then gain from production-side distortions. Under the linear technology maintained throughout, skill types are perfect substitutes,

relative wages cannot move, and production efficiency is automatic, so the results of this section are exact. If clean and dirty production employed different skill mixes, the constrained optimum would generically feature sector-differentiated prices and general-equilibrium incidence terms (Stiglitz 1982; Sachs et al. 2020).

7 Quantitative analysis

How large is the Mirrleesian correction to the carbon price, and how much welfare does it deliver over a flat carbon tax? I answer both by solving the model’s optimal mechanism globally, calibrated to US data, rather than evaluating the wedge formula at borrowed sufficient statistics. The solver extends the dual-costate recursion of Farhi and Werning (2013) and Golosov et al. (2016) to two consumption goods, a clean numéraire and a polluting energy good, and prices the externality as a constant marginal damage e entering the planner’s resource constraint, so the social producer price of energy is $1 + e$. This is the single-cohort reading of the social cost of carbon in Section 5: for a cohort too small to move the stock, the discounted damage weights Φ_j collapse to one constant per-unit cost, no pollution-stock state is carried, and the Pigouvian benchmark is $\tau^q = \text{SCC} = e$. The stock dynamics are a later phase. What the solved mechanism settles is the object a formula evaluation cannot: the *level* and the *incidence* of the optimal commodity wedge, and the welfare cost of implementing it with an anonymous instrument.

The section makes three points. The optimal carbon wedge is the social cost of carbon times a history-dependent Corlett–Hague premium: it starts at the Pigouvian level, rises to about nine percent above it by the end of the working life, and tracks the labor wedge, as the proportionality result of Section 4 predicts. The wedge is progressive in skill, so the regressivity of real carbon taxes is a property of flat instruments, not of the optimum. And a flat carbon tax at the social cost of carbon captures ninety-eight percent of the welfare of the full non-linear second best. The entire Mirrleesian fine structure of the carbon wedge is worth a few thousandths of a percent of consumption, which is the separation principle measured in welfare.

7.1 Model and calibration

The life-cycle block is Farhi and Werning (2013)’s, inherited from a validated replication. Agents live sixty years, work for forty and retire for twenty; log-skills follow a geometric random walk with innovation variance $\hat{\sigma}^2 = 0.0095$; the period consumption index enters utility logarithmically, and labor disutility is isoelastic with curvature $\alpha = 3$ (Frisch elasticity one half); $\beta = q = 0.95$; skills are private information and the incentive constraints are handled by the first-order approach. Each period the agent consumes a clean numéraire and an energy good (motor fuel, electricity, gas, fuel oil), both with producer price one. Energy is a relative leisure complement: its utility weight falls with labor effort at rate χ , so $U_{qn} \neq 0$ with the sign, in the notation of Section 4, that makes the complementarity gap $A_t = \gamma_c - \gamma_q$ positive. This is the empirically relevant case (West and Williams III (2007); Christiansen (1984)), and the one that pushes the optimal carbon tax above Pigouvian.

Two solvers deliver the results. The *second best* assigns full consumption bundles, so the commodity wedge is an unrestricted, history-dependent outcome; it requires homothetic preferences.

Linear commodity taxation layers an anonymous rate τ^q (flat or age-dependent) on the optimal non-linear income and savings mechanism, and reduces exactly to a one-good problem with reshaped labor disutility, valid with subsistence and hence usable for the necessity calibration. The two solvers agree where they must: at $A_t = 0$ the second best sets $\tau^q = e$ for every history and every age, the dynamic Atkinson–Stiglitz result, and it coincides with a flat Pigouvian tax to solver precision, returning the one-good initial condition $\hat{\lambda}_1 = 0.8797$. The incentive compatibility of the first-order-approach solution is verified ex post on the simulated benchmark.

The two-good calibration is disciplined by four US moments (Table 1). The energy expenditure share is seven percent (Consumer Expenditure Survey, 2023). The income elasticity of energy is 0.45, below one because energy is a necessity, from residential-energy demand studies; the homothetic benchmark raises it to one, and the Stone–Geary variant keeps it at 0.45. The own-price elasticity is 0.4, from meta-analyses of energy demand. The leisure complementarity χ is set so the zero-externality optimal energy tax matches West and Williams III (2007), about ten percent ad valorem for the household-energy composite. The marginal damage is $e = 0.494$: the EPA’s central social cost of carbon of \$190 per ton (U.S. Environmental Protection Agency 2023) times the carbon intensity of household energy spending, 0.0026 tons per producer dollar. Because $p_2 = 1$, the Pigouvian wedge $\tau^q = e$ is a forty-nine percent ad valorem tax on energy.

Table 1: Calibration

parameter	value	moment or source
<i>life-cycle block (Farhi and Werning 2013)</i>		
discount factor $\beta = q$	0.95	FW13
labor-disutility curvature α	3	Frisch elasticity 0.5
log-skill innovation variance $\hat{\sigma}^2$	0.0095	US male wages, via FW13
working / retirement periods	40/20	FW13
<i>two-good block</i>		
energy expenditure share	0.07	Consumer Expenditure Survey 2023
income elasticity of energy	0.45	residential-energy demand (necessity)
own-price elasticity of energy	0.4	energy-demand meta-analyses
leisure complementarity χ	0.25	zero-externality tax $\approx 10\%$ (West and Williams III 2007)
marginal damage e	0.494	\$190/tCO ₂ (U.S. Environmental Protection Agency 2023) $\times$ 0.0026 t

7.2 The optimal carbon wedge over the life cycle

The optimal wedge is the social cost of carbon plus a small, rising premium (Table 2, Figure 1). At labor-market entry the labor wedge is near zero, there is no incentive motive yet, and the carbon wedge sits at the Pigouvian level $e = 0.494$. It then rises concavely to about 0.63 near retirement. The non-Pigouvian premium, the markup over the Pigouvian gross price, tops

out around nine percent. So the optimal carbon tax exceeds the social cost of carbon, in the direction a leisure complement predicts, but by less than a tenth over most of the life cycle.

The premium is a scaled labor wedge. Its first-order value is $\chi n^\alpha \tau^y / (1 - \tau^y)$, and this identity tracks the solved premium within about ten percent at every age (last two rows of Table 2). The non-Pigouvian part of the carbon wedge therefore inherits the labor wedge's level and its life-cycle rise, which is the proportionality result of Section 4 made quantitative. The wedge is also genuinely history-dependent: its cross-sectional standard deviation fans out from a tenth of a percent early in life to five and a half percent near retirement, mirroring the spreading of the labor wedge as skill histories diverge.

The income-side wedges are essentially undisturbed by the second good. The labor wedge runs from 0.02 to 0.36 over the working life, against 0.02 to 0.37 in the one-good economy, and the savings wedge falls from 0.56 to 0.02 percent, the Farhi and Werning (2013) pattern intact. The commodity margin adds a corrective and screening instrument without reshaping the labor and savings margins, which is the separation showing up in the solved policies. Under the wedge the mean energy share falls from the calibrated seven percent to about six at producer prices, the households priced off energy at the calibrated elasticity.

Table 2: Cross-sectional mean wedges over the working life (benchmark, second best)

working year t	1	10	20	30	40
labor wedge τ^y	0.021	0.167	0.272	0.336	0.363
carbon wedge τ^q	0.504	0.569	0.606	0.624	0.632
s.d. across histories	0.001	0.005	0.014	0.029	0.055
non-Pigouvian premium	0.007	0.050	0.075	0.087	0.092
identity $\chi n^\alpha \tau^y / (1 - \tau^y)$	0.006	0.047	0.069	0.079	0.083

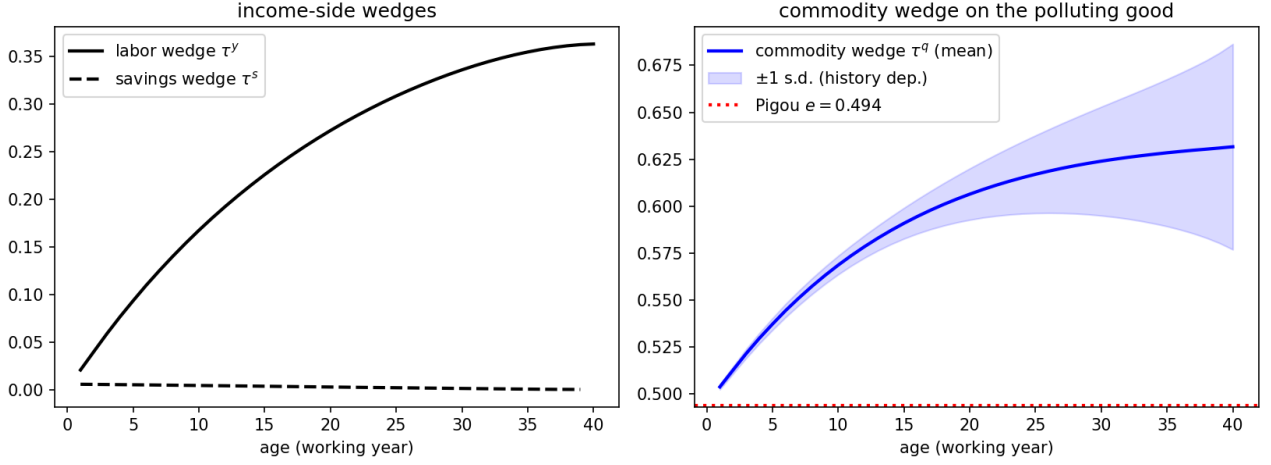


Figure 1: Income-side wedges (left) and the commodity wedge on energy (right), benchmark calibration. The band is ± 1 cross-sectional standard deviation across histories; the dotted line is the Pigouvian level $\tau^q = e = 0.494$.

7.3 Who bears the wedge

The optimal carbon wedge is progressive in skill (Figure 2). Binned by current productivity, the wedge is indistinguishable across quintiles early in life, when every history looks alike, and it fans out from mid-career: by the last working year it reaches about seventy percent for the top skill quintile against fifty-seven for the bottom, a fourteen-point spread. High current productivity signals a history whose incentive constraints are expensive, and the leisure-complement margin is the planner's lever on exactly those histories, so the screening value of the commodity instrument concentrates on high-skill agents.

Two readings follow. The optimal carbon wedge is not regressive: the planner taxes the energy of high-skill households more, not less, so the redistributive worry about carbon pricing is misplaced at the level of the optimal type-contingent wedge. The regressivity of real carbon taxes comes from elsewhere, from the flat-rate restriction, since an anonymous point-of-sale tax cannot condition on skill, and, in the necessity calibration, from energy's falling budget share. That is also why the flat rungs of the welfare ladder below leave money on the table.

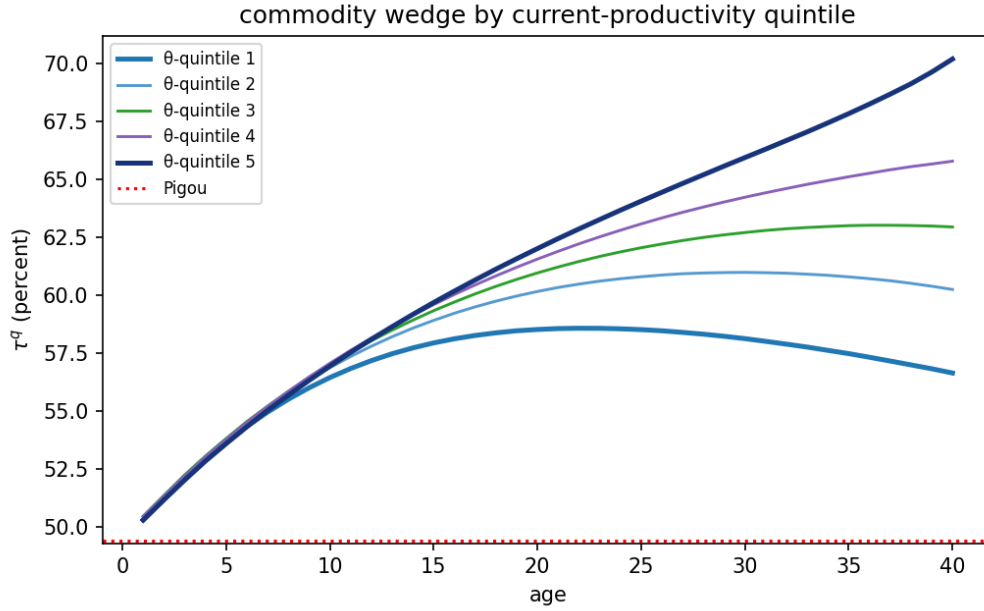


Figure 2: The optimal commodity wedge τ^q (percent) by current-skill quintile and age, benchmark calibration.

7.4 The welfare value of carbon pricing and the cost of simpler rules

Table 3 is the welfare accounting: consumption-equivalent gains over having no carbon price, under the optimal non-linear income and savings mechanism throughout, computed with common random numbers so that cross-regime differences net out simulation noise. Every regime pays for its own emissions in the planner’s budget, so the welfare differences are exact; the numerical resolution is about 0.002 percentage points. Six findings, in order of size.

The value of a carbon price is overwhelmingly Pigouvian. At the EPA social cost of carbon, having any carbon price is worth 0.31 percent of consumption, and a flat rate at the social cost of carbon alone captures ninety-eight percent of the full second-best value while cutting emissions by 13.5 percent. For scale, the entire insurance value of Farhi and Werning (2013)’s optimal income-tax apparatus over laissez-faire is 1.56 percent of consumption: the carbon instrument is worth about a fifth of that at this social cost of carbon.

The optimal flat rate exceeds the social cost of carbon by about fourteen percent, $\tau^{q*} = 0.56$ against $e = 0.494$, the flat reflection of the Corlett–Hague premium. But the welfare gain from that uplift over pricing at the social cost of carbon is three thousandths of a percent: the objective is very flat in the rate, so “price at the social cost of carbon” is a near-optimal rule of thumb even with non-separable preferences.

Sophistication buys almost nothing beyond the level. Making the rate age-dependent adds about 0.002 percentage points over the optimal flat rate, and full history dependence about 0.001 more, both at the numerical resolution. The premium varies over life and across histories by a few points of a six-percent budget share with a substitution elasticity of 0.4, so the deadweight loss from mistracking it is second-order squared.

The pure Corlett–Hague motive is tiny. With the externality off, the entire second-best commodity-wedge apparatus is worth 0.0036 percent of consumption, and its optimal flat rate is five percent, materially positive as in West and Williams III (2007) but negligible in welfare. Non-separability matters for the *level* of the optimal carbon tax, the fourteen-percent uplift, not as a stand-alone reason to tax energy.

Necessity halves the value and the abatement. Under the Stone–Geary calibration, in which energy’s budget share falls with income, the value of the flat carbon tax falls to 0.15 percent and its abatement to six percent, because the subsistence part of energy does not respond to price. The optimal flat rate is essentially unchanged. This is the calibration in which a flat carbon tax is regressive, yet the mechanism absorbs the incidence through the income side, which is why the flat instrument loses so little welfare despite its incidence.

The separation placebo. With the leisure complementarity switched off ($A_t = 0$), the second best and the flat Pigouvian tax coincide to five decimal places in welfare and in emissions: dynamic Atkinson–Stiglitz, and the exact knife-edge of the Corollary of Section 5, reproduced through two structurally different solvers.

Table 3: Welfare value of carbon pricing under progressively simpler rules (consumption-equivalent gain over $\tau^q = 0$)

	benchmark ($e = 0.494$)	Stone–Geary (necessity)	$A_t = 0$ (placebo)	no externality ($e = 0$)
second best (non-linear τ^q)	0.315	—	0.252	0.0036
age-dependent linear $\tau^q(t)$	0.314	—	—	0.0038
flat τ^{q*} (optimized)	0.313	0.153	—	0.0026
optimal flat rate τ^{q*}	0.56	0.56	—	0.051
flat Pigouvian $\tau^q = e$	0.309	0.151	0.252	—
emissions vs $\tau^q = 0$: second best	−15.0%	—	−13.5%	−1.8%
flat Pigouvian	−13.5%	−6.3%	−13.5%	−1.8%

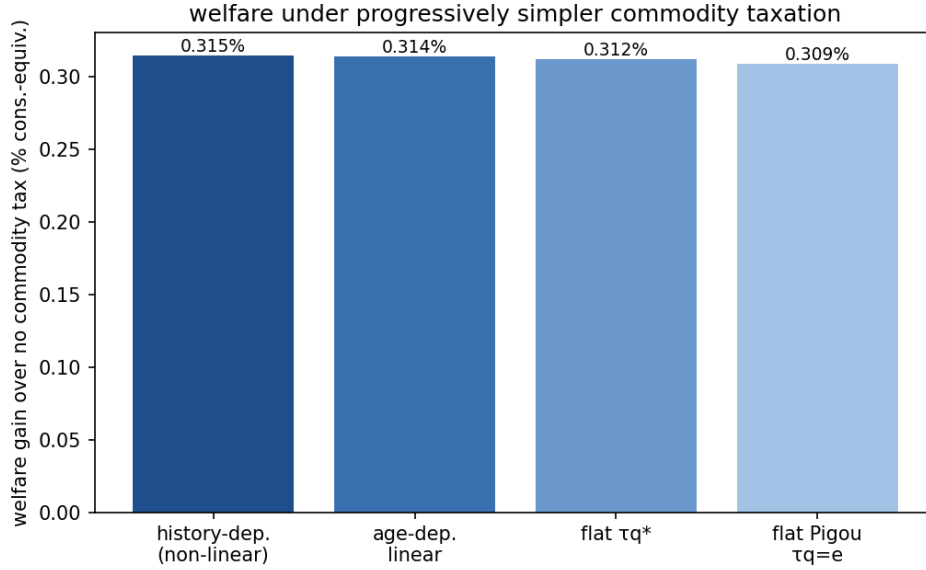


Figure 3: Consumption-equivalent welfare gain over no carbon price, benchmark calibration, under progressively simpler carbon-pricing rules.

7.5 Reading and limitations

The practical hierarchy for carbon policy in a Mirrleesian world is clean. Get the carbon price to the social cost of carbon, worth 0.31 percent of consumption; nudge it up by the leisure-complementarity premium, worth a further three thousandths; and spend the sophistication budget on the income tax, where Farhi and Werning (2013)’s own accounting shows the returns to age- and history-dependence are two orders of magnitude larger. The redistributive value embedded in the carbon tax is small, which is the separation principle in welfare terms: the income tax and the climate dividend, not the carbon price, should carry the distribution. This agrees with the near-Pigouvian conclusion of Douenne et al. (2026), reached here from the Mirrleesian rather than the Ramsey side, and it answers the distributional concern that opens the paper. Regressive carbon pricing and the backlash it provokes are best met by adjusting the income tax, not by tilting, means-testing, or weakening the carbon price.

Four limitations bound the numbers. The externality is a constant marginal damage, so the stock dynamics of Section 5, which move the *level* of the tax rather than the wedge structure measured here, are a separate exercise. The second best requires homothetic preferences, so the necessity calibration is read off the exact linear-tax reduction rather than the full non-linear mechanism. The log consumption index is required by the balanced-growth homogeneity the solver relies on. And the energy good carries a single carbon intensity, so within-energy heterogeneity is outside the model. None of these bears on the three results: the sign and

proportionality of the premium, its progressivity in skill, and the near-sufficiency of flat Pigouvian pricing are structural.

8 Conclusion

Using a dynamic Mirrlees framework, which features an explicit motive for income taxation, extended to feature consumption externalities, I find that the optimal pollution tax is only adjusted for inequality when there are non-separabilities in preferences between consumption, labor, and pollution. Otherwise, taxes are Pigouvian even in the presence of labor and capital income taxation, which shows that the dynamic Ramsey result that tax distortions justify deviating from Pigouvian taxation does not necessarily hold. Moreover, the optimal formulae for labor and savings wedges remain unchanged, suggesting that a properly designed commodity wedge is fully capable of internalizing the externality while respecting the social planner's redistributive goals.

When the commodity instrument must be a uniform anonymous price, the form real carbon taxes take, the two-good problem collapses back to a one-good dynamic Mirrlees problem in expenditure. The optimal uniform tax is a substitution-weighted average of the history-dependent wedges of the full optimum; the labor and savings margins pick up a fiscal-interaction term that leans the income tax against the commodity-tax base; and the welfare cost of the restriction is the substitution-weighted dispersion of the unconstrained wedge, which vanishes under weak separability. With the externality, the optimal uniform price splits additively into the social cost of carbon and a screening component, so the restriction binds only the redistributive part of the carbon price and never the Pigouvian part.

There is much room for further study. First, this paper focuses on wedges rather than tax implementation, so an interesting avenue of research may be assessing how the optimal commodity wedge may be replicated in practice. Second, while the body of the paper features a single generation, climate change is in practice very much an intergenerational problem. The natural next step is an overlapping-generations extension, in which the single-generation model becomes the component problem of an infinite-horizon planner and the fixed point between the carbon-price path and the per-cohort mechanism-design problems endogenizes the savings-wedge paths that section 7 takes from calibrated optima. Outside of the environmental context, since the optimal commodity wedge is a time-varying object in this dynamic model, one might also wonder if and how age-dependent commodity taxes may be used in a fiscal mix. Although age-dependent commodity taxation may at first glance appear a fanciful notion, positive age discrimination in the price of certain goods and services for young or old consumers or indeed discounts for low-income consumers already somewhat acts as implementing a time-varying income-based commodity wedge³.

³Of course, whether this is done for social or economic reasons is another question. However, the point is that it *can* be done. Future work could evaluate the possible deliberate use of age-dependent income-dependent commodity wedges to further efficiency or equity objectives.

References

- Barrage, Lint (2019). “Optimal Dynamic Carbon Taxes in a Climate–Economy Model with Distortionary Fiscal Policy”. In: *The Review of Economic Studies* 87.1, pp. 1–39. ISSN: 0034-6527. DOI: 10.1093/restud/rdz055.
- Bovenberg, A. Lans and Ruud A. de Mooij (1994). “Environmental Levies and Distortionary Taxation”. In: *The American Economic Review* 84.4, pp. 1085–1089. ISSN: 00028282. JSTOR: 2118046.
- Chancel, Lucas (2022). “Global carbon inequality over 1990–2019”. In: *Nature Sustainability* 5.11, pp. 931–938. DOI: 10.1038/s41893-022-00955-z.
- Chen, Yunmin and Dongmeng Ren (2024). “Consumption externalities, habit formation and optimal dynamic non-linear income taxation under asymmetric information”. In: *Pacific Economic Review* 29.4. Also appears as *Pacific Economic Review* 30(4), 2025; verify canonical version, pp. 505–529. DOI: 10.1111/1468-0106.12455.
- Christiansen, Vidar (1984). “Which commodity taxes should supplement the income tax?” In: *Journal of Public Economics* 24.2, pp. 195–220. ISSN: 0047-2727. DOI: [https://doi.org/10.1016/0047-2727\(84\)90024-0](https://doi.org/10.1016/0047-2727(84)90024-0). URL: <https://www.sciencedirect.com/science/article/pii/0047272784900240>.
- Crampes, Claude and Norbert Ladoux (2024). “Energy-Efficiency Investment and the Optimal Nonlinear Pricing of Energy”. In: *Journal of Public Economic Theory*.
- Cremer, Helmuth, Firouz Gahvari, and Norbert Ladoux (Dec. 1998). “Externalities and Optimal Taxation”. In: *Journal of Public Economics* 70.3, pp. 343–364. ISSN: 0047-2727. DOI: 10.1016/S0047-2727(98)00039-5.
- (2010). “Environmental tax design with endogenous earning abilities (with applications to France)”. In: *Journal of Environmental Economics and Management* 59.1, pp. 82–93.
- Cronin, Julie Anne, Don Fullerton, and Steven Sexton (2019). “Vertical and Horizontal Redistributions from a Carbon Tax and Rebate”. In: *Journal of the Association of Environmental and Resource Economists* 6.S1, pp. 169–208. DOI: 10.1086/701191.
- Diamond, Peter and James Mirrlees (1971). “Optimal Taxation and Public Production I: Production Efficiency”. In: *The American Economic Review* 61.1. Publisher: American Economic Association, pp. 8–27. ISSN: 00028282. URL: <http://www.jstor.org/stable/1910538> (visited on 06/23/2025).
- Dietsch, Peter (2024). “A fairer and more effective carbon tax”. In: *Nature Sustainability* 7.12, pp. 1584–1591. DOI: 10.1038/s41893-024-01429-0.
- Douenne, Thomas, Albert Jan Hummel, and Marcelo Pedroni (Jan. 2026). “Optimal Fiscal Policy in a Climate-Economy Model with Heterogeneous Households*”. In: *The Economic Journal*, ueag006. ISSN: 0013-0133. DOI: 10.1093/ej/ueag006. URL: <https://doi.org/10.1093/ej/ueag006> (visited on 04/19/2026).
- Douenne, Thomas et al. (2025). *Optimal Climate Policy with Incomplete Markets*. Tech. rep. Issue: tecipa-807. University of Toronto, Department of Economics. URL: <https://ideas.repec.org/p/tor/tecipa/tecipa-807.html>.

- Environment and Climate Change Canada (2023). *Social Cost of Greenhouse Gas Estimates – Interim Updated Guidance for the Government of Canada*.
- Farhi, Emmanuel and Ivan Werning (2013). “Insurance and Taxation over the Life Cycle”. In: *The Review of Economic Studies* 80.2, pp. 596–635. ISSN: 0034-6527. DOI: 10.1093/restud/rds048.
- Golosov, Mikhail, Narayana Kocherlakota, and Aleh Tsyvinski (2003). “Optimal Indirect and Capital Taxation”. In: *Review of Economic Studies* 70.3, pp. 569–587. ISSN: 0034-6527. DOI: 10.1111/1467-937x.00256.
- Golosov, Mikhail, Maxim Troshkin, and Aleh Tsyvinski (2016). “Redistribution and Social Insurance”. In: *American Economic Review* 106.2, pp. 359–86. DOI: 10.1257/aer.20111550.
- Golosov, Mikhail et al. (2014). “Optimal Taxes on Fossil Fuel in General Equilibrium”. In: *Econometrica* 82.1, pp. 41–88. ISSN: 0012-9682. DOI: 10.3982/ECTA10217.
- Hänsel, Martin C. et al. (2022). “Optimal Carbon Taxation and Horizontal Equity: A Welfare-Theoretic Approach with Application to German Household Data”. In: *Journal of Environmental Economics and Management* 116, p. 102730. DOI: 10.1016/j.jeem.2022.102730.
- Hassett, Kevin A., Aparna Mathur, and Gilbert E. Metcalf (2009). “The Incidence of a U.S. Carbon Tax: A Lifetime and Regional Analysis”. In: *The Energy Journal* 30.2, pp. 155–177. DOI: 10.5547/ISSN0195-6574-EJ-Vol130-No2-8.
- Hellwig, Christian (2021). *Static and Dynamic Mirrleesian Taxation with Non-separable Preferences: A Unified Approach*. Working Paper 1224. Toulouse School of Economics.
- Interagency Working Group on Social Cost of Greenhouse Gases (2021). *Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide Interim Estimates under Executive Order 13990*.
- Jacobs, Bas and Ruud A. de Mooij (2015). “Pigou Meets Mirrlees: On the Irrelevance of Tax Distortions for the Second-Best Pigouvian Tax”. In: *Journal of Environmental Economics and Management* 71, pp. 90–108. ISSN: 0095-0696. DOI: 10.1016/j.jeem.2015.01.003.
- Jacobs, Bas and Frederick van der Ploeg (2019). “Redistribution and Pollution Taxes with Non-Linear Engel Curves”. In: *Journal of Environmental Economics and Management* 95, pp. 198–226. ISSN: 0095-0696. DOI: 10.1016/j.jeem.2019.01.008.
- Känzig, Diego R. (2023). “The Unequal Economic Consequences of Carbon Pricing”. In: *NBER Working Paper 31221*. Forthcoming, American Economic Review. DOI: 10.3386/w31221.
- Kapička, Marek (2013). “Efficient Allocations in Dynamic Private Information Economies with Persistent Shocks: A First-Order Approach”. In: *The Review of Economic Studies* 80.3, pp. 1027–1054. ISSN: 0034-6527. DOI: 10.1093/restud/rds045.
- Kaplow, Louis (2012). “Optimal Control of Externalities in the Presence of Income Taxation”. In: *International Economic Review* 53.2, pp. 487–509. ISSN: 0020-6598. DOI: 10.1111/j.1468-2354.2012.00689.x.
- Naito, Hisahiro (1999). “Re-examination of uniform commodity taxes under a non-linear income tax system and its implication for production efficiency”. In: *Journal of Public Economics* 71.2, pp. 165–188.

- Parry, Ian W. H., Roberton C. Williams III, and Lawrence H. Goulder (1999). “When Can Carbon Abatement Policies Increase Welfare? The Fundamental Role of Distorted Factor Markets”. In: *Journal of Environmental Economics and Management* 37.1, pp. 52–84. DOI: 10.1006/jeem.1998.1058.
- Pizer, William A. and Steven Sexton (2019). “The Distributional Impacts of Energy Taxes”. In: *Review of Environmental Economics and Policy* 13.1, pp. 104–123. DOI: 10.1093/reep/rey021.
- Poterba, James M. (1989). “Lifetime Incidence and the Distributional Burden of Excise Taxes”. In: *The American Economic Review* 79.2. Papers and Proceedings, pp. 325–330.
- Sachs, Dominik, Aleh Tsyvinski, and Nicolas Werquin (2020). “Nonlinear Tax Incidence and Optimal Taxation in General Equilibrium”. In: *Econometrica* 88.2, pp. 469–493. DOI: 10.3982/ECTA14681.
- Sandmo, Agnar (1975). “Optimal Taxation in the Presence of Externalities”. In: *The Swedish Journal of Economics* 77.1, pp. 86–98. ISSN: 00397318. DOI: 10.2307/3439329. JSTOR: 3439329.
- Stiglitz, Joseph E. (1982). “Self-selection and Pareto efficient taxation”. In: *Journal of Public Economics* 17.2, pp. 213–240.
- U.S. Environmental Protection Agency (2023). *Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances*. Report. National Center for Environmental Economics, Office of Policy.
- van der Ploeg, Frederick, Armon Rezai, and Miguel Tovar Reaños (2025). “Third-Best Carbon Taxation: Trading off Emission Cuts, Equity, and Efficiency”. In: *Journal of the Association of Environmental and Resource Economists*. CEPR Discussion Paper DP19612.
- Weinzierl, Matthew (2011). “The Surprising Power of Age-Dependent Taxes”. In: *The Review of Economic Studies* 78.4, pp. 1490–1518.
- West, Sarah E. and Roberton C. Williams III (2007). “Optimal Taxation and Cross-Price Effects on Labor Supply: Estimates of the Optimal Gas Tax”. In: *Journal of Public Economics* 91.3–4, pp. 593–617. ISSN: 0047-2727. DOI: 10.1016/j.jpubeco.2006.08.012.

A Optimal wedge derivation without externalities

Because $U(c, q, l)$ can be written as $U(c, q, y/\theta)$, I can write U_θ as $-U_l l/\theta$. The Hamiltonian to the recursive problem is

$$\begin{aligned}\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta) \right] f \\ & + \psi \left[-U_l(c, q, y/\theta) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) \bar{\alpha}_t f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(c, q, y/\theta) - \beta w],\end{aligned}$$

Differentiating yields the following first-order conditions

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial c} : & f - \psi U_{cl} \frac{l}{\theta} = \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} : & f - \psi U_{ql} \frac{l}{\theta} = \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} : & -\theta f - \phi U_l = \frac{1}{\theta} \psi [U_{ll} l + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} : & \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\ \frac{\partial \mathcal{H}}{\partial w} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta.\end{aligned}$$

A.1 Labor wedge

Dividing the first-order condition for c by U_c and using $\gamma_c \equiv U_{cl} l/U_c$,

$$\phi = \frac{f}{U_c} - \frac{\psi}{\theta} \gamma_c. \tag{1}$$

From the first-order condition for l , and since $U_{ll} l + U_l = U_l(1 + \epsilon)$ with $\epsilon \equiv U_{ll} l/U_l$,

$$-\theta f - \phi U_l = \frac{\psi}{\theta} U_l (1 + \epsilon) \implies \phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta} (1 + \epsilon).$$

Equating the two expressions for ϕ and collecting the terms in ψ ,

$$\frac{f}{U_c} + \frac{\theta f}{U_l} = -\frac{\psi}{\theta} (1 + \epsilon - \gamma_c).$$

The labor wedge is $1 - \tau^y = -U_l/(\theta U_c)$, so we can write $\theta U_c/U_l = -1/(1 - \tau^y)$, making the left-hand side $\frac{f}{U_c} (1 - \frac{1}{1 - \tau^y}) = -\frac{f}{U_c} \frac{\tau^y}{1 - \tau^y}$. Substituting and rearranging,

$$\frac{\tau^y}{1 - \tau^y} = (1 + \epsilon - \gamma_c) \frac{\psi U_c}{\theta f} = A_t(\theta) \Lambda_t(\theta) \quad (2)$$

Next, substitute ϕ from (1) into $\dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi$,

$$\dot{\psi} - \frac{\gamma_c}{\theta} \psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}.$$

This is a linear first-order ordinary differential equation. Multiplying by the integrating factor $\exp \left[-\int^\theta \gamma_c(\tilde{x}) d\tilde{x}/\tilde{x} \right]$ and integrating from θ to ∞ with the boundary condition $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp \left[-\int_\theta^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx. \quad (3)$$

The remaining boundary condition $\psi(0) = 0$ pins down λ_1 ,

$$\lambda_{1,t} = \frac{\int_0^\infty \exp \left[-\int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_0^\infty \exp \left[-\int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \bar{\alpha}_t(x) f_t(x) dx}.$$

Substituting (3) into (2),

$$\frac{\tau^y}{1 - \tau^y} = (1 + \epsilon - \gamma_c) \frac{U_c(\theta)}{\theta f(\theta)} \int_\theta^\infty \exp \left[-\int_\theta^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx.$$

I split the integral into the $[f/U_c - \lambda_1 \bar{\alpha}_t f]$ part, which yields the intratemporal term $A_t B_t C_t$, and the $\lambda_2 f_2$ part, which yields the intertemporal term. For the first part, carry $U_c(\theta)/U_c(x)$ into the integrand. Along the optimal allocation $U_{c,t}$ depends on the type through $c(\theta)$, $q(\theta)$, and $l(\theta) = y(\theta)/\theta$, so

$$\frac{d}{d\theta} \ln U_{c,t} = \frac{U_{cc}\dot{c} + U_{cq}\dot{q} + U_{cl}\dot{l}}{U_c} = -\sigma_t \frac{\dot{c}}{c} - \sigma_{cq,t} \frac{\dot{q}}{q} + \gamma_c \frac{\dot{l}}{l}.$$

Since $l = y/\theta$ gives $\dot{l}/l = \dot{y}/y - 1/\theta$, integrating from θ to x and multiplying by the integrating factor gives the identity

$$\frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] = \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_c(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right],$$

in which the $\gamma_c/\tilde{x}$ term from $\dot{l}/l$ cancels the integrating factor. The cross-curvature $\sigma_{cq,t}$ enters because, under non-separability, extracting resources from a higher type shifts the marginal utility of the numéraire through the adjustment of both goods. The first part therefore equals $A_t B_t C_t$. For the second part, the first-order condition for w_2 gives $\partial V_{t+1}/\partial w_2 = -\beta R \psi/f$, and the envelope condition $\partial V_t/\partial \hat{w}_2 = -\lambda_{2,t}$ one period ahead gives $\lambda_{2,t}(\theta^{t-1}) = \beta R \psi_{t-1}/f_{t-1}$. Writing $\psi_{t-1}/f_{t-1} = \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} \frac{\theta_{t-1}}{U_{c,t-1} A_{t-1}}$ from (2) at $t-1$ and applying the same integrating-factor manipulation to $\lambda_2 f_2$ produces $\beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} D_t(\theta)$. Collecting both parts,

$$\frac{\tau^y}{1-\tau^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} D_t(\theta),$$

$$A_t(\theta) = 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta) U_{c,t}(\theta) \theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{A_{t-1} U_{c,t-1} \theta f_t(\theta)}.$$

This is Golosov et al. (2016)'s labor-wedge decomposition, with the cross-curvature $\sigma_{cq,t}$ in C_t .

A.2 Savings wedge

The savings wedge follows from the same first-order conditions. Define

$$\chi_t(\theta^t) \equiv \frac{\gamma_{c,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y},$$

Using $\frac{\tau_t^y}{1 - \tau_t^y} = \frac{\psi U_c}{\theta f} (1 + \epsilon_t - \gamma_{c,t})$, $\phi = f/U_c - \psi \gamma_c / \theta$ from (1) can be rewritten as follows:

$$\frac{\phi}{f} = \frac{1}{U_c} (1 - \gamma_c \psi U_c / (\theta f)) = \frac{1 - \chi_t}{U_{c,t}}. \quad (4)$$

χ_t is positive when the numéraire is complementary with labor ($U_{cl} > 0$) and labor is taxed, and is zero when preferences are separable in the numéraire against labor ($\gamma_c = 0$), or when the labor margin is undistorted. The first-order condition for w and the envelope condition $\partial V_t / \partial \hat{w} = \lambda_{1,t}$ one period ahead give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1 - \chi_t}{U_{c,t}}. \quad (5)$$

A second expression for $\lambda_{1,t}$ comes from integrating the costate equation over the whole type space. With $\psi(0) = \psi(\infty) = 0$,

$$0 = \int_0^\infty \dot{\psi} d\theta = \lambda_{1,t} \int_0^\infty \bar{\alpha}_t f d\theta - \lambda_{2,t} \int_0^\infty f_2 d\theta - \int_0^\infty \phi d\theta.$$

The Pareto weights are normalized so that $\int \bar{\alpha}_t f d\theta = 1$, and $\int f_t(\theta | \theta_-) d\theta = 1$ for every θ_- implies $\int f_{2,t} d\theta = 0$. Hence $\lambda_{1,t} = \int_0^\infty \phi d\theta$, which by (4) is

$$\lambda_{1,t} = \mathbb{E}_t \left[\frac{1 - \chi_t}{U_{c,t}} \right]. \quad (6)$$

Evaluating (6) at $t + 1$ and equating with (5),

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right].$$

Defining the incentive-adjusted marginal utility $\widehat{U}_{c,t} \equiv U_{c,t}/(1 - \chi_t)$,

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right].$$

This is the classical inverse Euler equation in $\widehat{U}_c$ where non-separability merely adjusts the marginal utility of the numéraire. Delivering a util to type θ tomorrow costs $\frac{1}{\widehat{U}_{c,t}}$ today, but a marginal unit of c also shifts the labor a local mimicker would exert at shadow value $\chi_t/U_{c,t}$. This is the generalized inverse Euler condition like the one derived in Hellwig 2021.

A.3 Commodity wedge

The commodity wedge uses the first-order conditions for c and q of the preamble. There are different equivalent ways to express it, but I will seek the familiar labor wedge $ABCD$ form. First, because the first-order conditions for each good share the same ϕ ,

$$\frac{f_t}{U_q} - \psi \frac{U_{ql}}{U_q} \frac{l}{\theta} = \frac{f_t}{U_c} - \psi \frac{U_{cl}}{U_c} \frac{l}{\theta}$$

If we cross-multiply by U_q and U_c and use $\gamma_c \equiv \frac{U_{cl}l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql}l}{U_q}$,

$$U_c \left(f - \frac{\psi}{\theta} \gamma_q U_q \right) = U_q \left(f - \frac{\psi}{\theta} \gamma_c U_c \right)$$

Then, if we divide by $U_c f$ and rearrange terms,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f} (\gamma_q - \gamma_c)$$

Note that, by definition, $1 + \tau^q = \frac{U_q}{U_c}$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1 + \tau^q}$. Multiplying through by -1 ,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q)$$

From here, using the results from the derivation of the labor wedge,

$$\frac{\tau^q}{1 + \tau^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t C_t + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{D_t}{A_t} \right],$$

with A_t, B_t, C_t, D_t as in Appendix A.1. Alternatively,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_t} \frac{\tau_t^y}{1 - \tau_t^y}$$

The commodity wedge inherits every dynamic feature of the labor wedge: its shape is proportional to that of the labor wedge, scaled by the elasticity term.

B Optimal wedge derivation with externalities

B.1 Commodity wedge

The Hamiltonian to this problem is as follows.

$$\begin{aligned}\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, S) \right] f \\ & + \psi \left[-U_l(c, q, y/\theta, S) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) \bar{\alpha}_t f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(c, q, y/\theta, S) - \beta w] \\ & - \mu [S - S(\dots, Q)]\end{aligned}$$

This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial c} : & f - \psi U_{cl} \frac{l}{\theta} = \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} : & f - \psi U_{ql} \frac{l}{\theta} + \mu S_q f = \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} : & -\theta f - \phi U_l = \frac{1}{\theta} \psi [U_{ll} l + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} : & \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\ \frac{\partial \mathcal{H}}{\partial w} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta \\ \frac{\partial \mathcal{H}}{\partial S} : & \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f - \psi U_{ls} \frac{l}{\theta} - \phi U_s \right) d\theta = \mu\end{aligned}$$

As in the previous proof,

$$\phi = \frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c}$$

Because c and q share the same ϕ ,

$$\phi = \frac{f}{U_q} - \frac{\psi U_{ql} \frac{l}{\theta}}{U_q} + \frac{\mu S_q f}{U_q}$$

Hence,

$$\frac{f}{U_q} - \frac{\psi U_{ql} \frac{l}{\theta}}{U_q} + \frac{\mu S_q f}{U_q} = \frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c}$$

If we multiply by U_q and U_c ,

$$U_c \left(f - \psi U_{ql} \frac{l}{\theta} + \mu S_q f \right) = U_q \left(f - \psi U_{cl} \frac{l}{\theta} \right)$$

Using $\gamma_c \equiv \frac{U_{cl} l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql} l}{U_q}$, or, equivalently, $\gamma_c U_c = U_{cl} l$ and $\gamma_q U_q = U_{ql} l$,

$$U_c \left(f - \frac{\psi}{\theta} \gamma_q U_q + \mu S_q f \right) = U_q \left(f - \frac{\psi}{\theta} \gamma_c U_c \right)$$

Then, putting the γ terms (as well as the externality term) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta} U_c U_q (\gamma_q - \gamma_c) - \mu U_c S_q f$$

Finally, dividing by $U_q f$,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f} (\gamma_q - \gamma_c) - \mu \frac{U_c}{U_q} S_q$$

Note that, by definition, $1 + \tau^q = U_q/U_c$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1 + \tau^q}$. Multiplying through by -1 ,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \mu \frac{U_c}{U_q} S_q$$

Since $\frac{U_c}{U_q} = \frac{1}{1+\tau^q}$,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \frac{\mu S_q}{1+\tau^q}$$

If we rearrange terms,

$$\frac{\tau^q - \mu S_q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$$

The right-hand side is exactly that of the no-pollution primitive condition $\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$ of appendix A.3, with τ^q replaced by the deviation $\tau^q - \mu S_q$. The first-order conditions for c and l and the costate equation for ψ are untouched by the externality, so the integrating-factor steps there carry over verbatim and give

$$\frac{\tau_t^q - \mu_t S_q}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1-\tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right],$$

with A_t, B_t, C_t, D_t the labor-wedge terms of appendix A.1 (the cross-curvature $\sigma_{cq,t}$ included in C_t). The externality leaves this block untouched and enters only by replacing τ^q with the deviation $\tau^q - \mu S_q$ on the left.

Now let us focus on μS_q .

$$\int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f - \psi U_{ls} \frac{l}{\theta} - \phi U_S \right) S_q$$

If we substitute ϕ .

$$\int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f - \psi U_{ls} \frac{l}{\theta} - \left[\frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c} \right] U_S \right) S_q = \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \frac{\psi}{\theta f} U_{ls} l - \left[1 - \frac{\psi}{\theta f} U_{cl} l \right] \frac{U_S}{U_c} \right) S_q f$$

We can rewrite $\psi/\theta f$ noting that

$$\frac{\psi}{\theta f} = \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{U_c} \frac{1}{1 + \epsilon - \gamma_c}$$

Hence, also defining $\xi \equiv \frac{U_{ls} l}{U_S}$,

$$\begin{aligned} \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \frac{\psi}{\theta f} U_{ls} l - \left[1 - \frac{\psi}{\theta f} U_{cl} l \right] \frac{U_S}{U_c} \right) S_q f &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{ls} l}{U_c} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{U_{cl} l}{U_c} \right] \frac{U_S}{U_c} \right) S_q f \\ &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{ls} l}{U_c} \frac{U_S}{U_S} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{U_{cl} l}{U_c} \right] \frac{U_S}{U_c} \right) S_q f \\ &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi}{1 + \epsilon - \gamma_c} \frac{U_S}{U_c} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{U_{cl} l}{U_c} \right] \frac{U_S}{U_c} \right) S_q f \\ &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left[1 + \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi - \gamma_c}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) S_q f \end{aligned}$$

By the envelope theorem applied to the period- $(t + 1)$ problem, the lagged stock affects continuation costs only through the accumulation constraint, so $\frac{\partial V_{t+1}}{\partial S}$ collects μ_{t+1} times the marginal persistence of the stock. Iterating forward, each future term passes through the objective's $\frac{1}{R} V_{t+1}$ once per period and is therefore discounted at the technology rate. Defining $\Phi_j \equiv \partial S_{t+j} / \partial Q_t$ ($\Phi_j = 0$ for $j > s$ by assumption),

$$\text{SCC}_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + \left(\frac{\tau_{t+j}^y}{1 - \tau_{t+j}^y} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

Here, s is the number of periods emissions affect the pollution stock.

C Uniform commodity wedge derivations

Throughout this appendix I write $\rho \equiv 1 + T$ for the consumer price of the dirty good (producer price one, uniform tax T), and I keep the multiplier convention of appendix A.1: ψ is the costate on the envelope condition, ϕ the multiplier on the pointwise definition of u , and λ_1, λ_2 the multipliers on promise and threat keeping.

C.1 The inner problem and demand system

The agent splits assigned expenditure x at the common price ρ ,

$$\tilde{U}(x, l; T) = \max_{c, q} U(c, q, l) \quad \text{s.t.} \quad c + \rho q = x,$$

with the interior split characterized by the tangency $U_q = \rho U_c$ and the budget, defining $c^d(x, l, T)$ and $q^d(x, l, T)$.

Envelope properties. Differentiate $\tilde{U} = U(c^d, q^d, l)$. Differentiating the budget in x gives $\partial_x c^d + \rho \partial_x q^d = 1$, so, using $U_q = \rho U_c$,

$$\tilde{U}_x = U_c \partial_x c^d + U_q \partial_x q^d = U_c (\partial_x c^d + \rho \partial_x q^d) = U_c.$$

Differentiating the budget in l at fixed x gives $\partial_l c^d + \rho \partial_l q^d = 0$, so

$$\tilde{U}_l = U_l + U_c (\partial_l c^d + \rho \partial_l q^d) = U_l.$$

Differentiating the budget in T at fixed x (recall $\partial \rho / \partial T = 1$) gives $\partial_T c^d + q^d + \rho \partial_T q^d = 0$, so

$$\tilde{U}_T = U_c (\partial_T c^d + \rho \partial_T q^d) = -q^d U_c,$$

which is Roy's identity. This proves the first claim of Lemma 6.1.1. The second claim, that incentive compatibility in the restricted economy is equivalent to incentive compatibility in the one-good economy of appendix A.1 with period utility $\tilde{U}(x, y/\theta; T)$, follows because a deviator who reports $\hat{\theta}$ receives $(x(\hat{\theta}), y(\hat{\theta}))$, works $l = y(\hat{\theta})/\theta$, and splits $x(\hat{\theta})$ optimally at the same common price; the deviation space is therefore that of a one-good economy in $\tilde{U}$, and the type enters $\tilde{U}(x, y/\theta; T)$ only through $l = y/\theta$, so the envelope condition is $\dot{u} = -\tilde{U}_l l/\theta + \beta w_2$ as in appendix A.1.

Demand derivatives. Let $\mathcal{D} \equiv U_{qq} - 2\rho U_{cq} + \rho^2 U_{cc} < 0$ (strict concavity of U along the budget line). Differentiating the tangency $U_q = \rho U_c$ together with the budget:

In x at fixed l, ρ . The budget gives $\partial_x c^d = 1 - \rho \partial_x q^d$. Substituting into the differentiated tangency $U_{qc} \partial_x c^d + U_{qq} \partial_x q^d = \rho(U_{cc} \partial_x c^d + U_{cq} \partial_x q^d)$ and collecting $\partial_x q^d$,

$$(U_{qq} - 2\rho U_{cq} + \rho^2 U_{cc}) \partial_x q^d = U_{cq} - \rho U_{cc} \quad \implies \quad \partial_x q^d = \frac{U_{cq} - \rho U_{cc}}{|\mathcal{D}|}, \quad m_q \equiv \rho \partial_x q^d.$$

In l at fixed x, ρ . The budget gives $\partial_l c^d = -\rho \partial_l q^d$, and the differentiated tangency yields $\mathcal{D} \partial_l q^d = \rho U_{cl} - U_{ql}$. Using $U_{cl} = \gamma_c U_c / l$ and $U_{ql} = \gamma_q U_q / l = \gamma_q \rho U_c / l$,

$$\rho U_{cl} - U_{ql} = \frac{\rho U_c}{l} (\gamma_c - \gamma_q) \quad \implies \quad \partial_l q^d = \frac{\rho U_c (\gamma_q - \gamma_c)}{l |\mathcal{D}|},$$

where the sign flips because $\mathcal{D} < 0$.

Compensated own-price response. Along an indifference curve $dc = -\rho dq$; differentiating the tangency in ρ at constant utility gives $\mathcal{D} d\rho = U_c d\rho$, hence

$$s_{qq} \equiv \left. \frac{\partial q^d}{\partial \rho} \right|_u = \frac{U_c}{\mathcal{D}} = -\frac{U_c}{|\mathcal{D}|} < 0, \quad \frac{\partial q^d}{\partial \rho} = s_{qq} - q^d \partial_x q^d \quad (\text{Slutsky}).$$

C.2 Composite statistics

Composite complementarity. Since $\tilde{U}_x = U_c(c^d, q^d, l)$ and $\partial_l c^d = -\rho \partial_l q^d$,

$$\tilde{U}_{xl} = U_{cc} \partial_l c^d + U_{cq} \partial_l q^d + U_{cl} = U_{cl} + (U_{cq} - \rho U_{cc}) \partial_l q^d.$$

Dividing by $\tilde{U}_x = U_c$, multiplying by l , and substituting the demand derivatives,

$$\tilde{\gamma} = \frac{l \tilde{U}_{xl}}{U_c} = \gamma_c + \frac{l (U_{cq} - \rho U_{cc})}{U_c} \cdot \frac{\rho U_c (\gamma_q - \gamma_c)}{l |\mathcal{D}|} = \gamma_c + m_q (\gamma_q - \gamma_c) = (1 - m_q) \gamma_c + m_q \gamma_q.$$

Le Chatelier inequality. Since $\tilde{U}_l = U_l(c^d, q^d, l)$,

$$\tilde{U}_l = U_l + (U_{lq} - \rho U_{lc}) \partial_l q^d = U_l + \frac{\rho U_c(\gamma_q - \gamma_c)}{l} \cdot \frac{\rho U_c(\gamma_q - \gamma_c)}{l |\mathcal{D}|} = U_l + \frac{\rho^2 U_c^2 (\gamma_c - \gamma_q)^2}{l^2 |\mathcal{D}|},$$

using $U_{lq} - \rho U_{lc} = (\rho U_c/l)(\gamma_q - \gamma_c)$ from the demand step. Dividing by $\tilde{U}_l = U_l < 0$ flips the sign of the positive correction,

$$\tilde{\epsilon} = \frac{\tilde{U}_l l}{U_l} = \epsilon - \frac{\rho^2 U_c^2 (\gamma_c - \gamma_q)^2}{l |U_l| |\mathcal{D}|} \leq \epsilon.$$

Under weak separability $\gamma_c = \gamma_q$ (appendix A.3), so both corrections vanish and the reduced problem is that of appendix A.1 exactly.

C.3 Planner problem and first-order conditions

With $l = y/\theta$ as the control and the revenue-inclusive resource cost $x - Tq^d$, the recursive problem is that of appendix A.1 in the assigned pair (x, y) . Its Hamiltonian is

$$\begin{aligned} \mathcal{H} = & \left[x - Tq^d(x, l, T) - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta) \right] f \\ & + \psi \left[-\tilde{U}_l \frac{l}{\theta} + \beta w_2 \right] - \lambda_1 \bar{\alpha}_t u f + \lambda_2 u f_2 + \phi \left[u - \tilde{U} - \beta w \right]. \end{aligned}$$

Differentiating, and using $l \tilde{U}_{xl} = \tilde{\gamma} \tilde{U}_x$ and $\tilde{U}_l l + \tilde{U}_l = \tilde{U}_l(1 + \tilde{\epsilon})$,

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial x} : (1 - T \partial_x q^d) f - \psi \tilde{U}_{xl} \frac{l}{\theta} = \phi \tilde{U}_x & \iff \phi = \frac{(1 - T \partial_x q^d) f}{\tilde{U}_x} - \psi \frac{\tilde{\gamma}}{\theta} \\ \frac{\partial \mathcal{H}}{\partial l} : -(\theta + T \partial_l q^d) f - \phi \tilde{U}_l = \frac{1}{\theta} \psi \tilde{U}_l (1 + \tilde{\epsilon}) & \iff \phi = \frac{(\theta + T \partial_l q^d) f}{-\tilde{U}_l} - \psi \frac{1 + \tilde{\epsilon}}{\theta} \\ \frac{\partial \mathcal{H}}{\partial w} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \phi \beta & \iff \lambda_{1,t+1} = \beta R \frac{\phi}{f} \\ \frac{\partial \mathcal{H}}{\partial w_2} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta & \iff \lambda_{2,t+1} = \beta R \frac{\psi}{f} \\ \frac{\partial \mathcal{H}}{\partial u} : \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi, & \psi(0) = \psi(\infty) = 0. \end{aligned}$$

Relative to appendix A.1, the first two conditions carry the fiscal terms $T\partial_x q^d$ and $T\partial_l q^d$: assigning a unit of expenditure returns $T\partial_x q^d$ as commodity revenue, and assigning labor moves demand for the taxed good by $\partial_l q^d$. Define, as before, $\tilde{\Lambda}_t \equiv \psi \tilde{U}_x / (\theta f)$, and recall $1 - \tau^y = -\tilde{U}_l / (\theta \tilde{U}_x)$.

C.4 Labor wedge (Result 6.3.1)

Equate the two expressions for ϕ and collect the ψ terms,

$$\frac{(1 - T\partial_x q^d)f}{\tilde{U}_x} - \frac{(\theta + T\partial_l q^d)f}{-\tilde{U}_l} = \frac{\psi}{\theta} (\tilde{\gamma} - 1 - \tilde{\epsilon}) = -\frac{\psi}{\theta} \tilde{A}.$$

Multiply by $\tilde{U}_x/f$ and use $\tilde{U}_x/(-\tilde{U}_l) = 1/(\theta(1 - \tau^y))$,

$$1 - T\partial_x q^d - \frac{\theta + T\partial_l q^d}{\theta(1 - \tau^y)} = -\frac{\psi \tilde{U}_x}{\theta f} \tilde{A} = -\tilde{A} \tilde{\Lambda}.$$

Since $1 - 1/(1 - \tau^y) = -\tau^y/(1 - \tau^y)$, the left-hand side is

$$-\frac{\tau^y}{1 - \tau^y} - T\partial_x q^d - \frac{T\partial_l q^d}{\theta(1 - \tau^y)} = -\frac{\tau^y}{1 - \tau^y} - \frac{T}{1 - \tau^y} \left[(1 - \tau^y)\partial_x q^d + \frac{\partial_l q^d}{\theta} \right].$$

Recognizing the bracket as $dq^d/dy|_u$ and rearranging gives Result 6.3.1,

$$\frac{\tau^y}{1 - \tau^y} + \frac{T}{1 - \tau^y} \frac{dq^d}{dy} \Big|_u = \tilde{A} \tilde{\Lambda}.$$

The compensation reading of the bracket: along a perturbation dy with $d\tilde{U} = \tilde{U}_x dx + \tilde{U}_l dy/\theta = 0$, i.e. $dx = (1 - \tau^y)dy$, demand changes by $dq^d = \partial_x q^d dx + \partial_l q^d dy/\theta = [(1 - \tau^y)\partial_x q^d + \partial_l q^d/\theta] dy$.

C.5 Optimal uniform commodity tax (Result 6.3.2)

The uniform tax T_t is a scalar per period, common to the whole date- t cross-section. It enters the date- t problem in three places; differentiate each.

(i) *Resource cost.* $\partial_T[x - Tq^d(x, l, T)] = -(q^d + T\partial_T q^d)$, using $\partial_T T = 1$ and writing $\partial_T q^d = \partial q^d/\partial \rho$.

(ii) *Period utility*, carried by ϕ in $u = \tilde{U} + \beta w$: contribution $-\phi \tilde{U}_T = +\phi q^d \tilde{U}_x$ by Roy.

(iii) *The envelope constraint*, carried by ψ : the bracket contains $-\tilde{U}_l l/\theta$, and by Clairaut symmetry $\tilde{U}_{lT} = \partial_l(-q^d \tilde{U}_x) = -\partial_l q^d \tilde{U}_x - q^d \tilde{U}_{xl}$, so the contribution is $-\psi_\theta^l \tilde{U}_{lT} = +\psi_\theta^l (\partial_l q^d \tilde{U}_x + q^d \tilde{U}_{xl})$.

Sum per unit density. Substitute $\phi/f = (1 - T\partial_x q^d - \tilde{\chi})/\tilde{U}_x$ with $\tilde{\chi} \equiv \tilde{\gamma}\tilde{\Lambda}$ (from the FOC for x), $\psi\tilde{U}_x/(\theta f) = \tilde{\Lambda}$, and $l\tilde{U}_{xl} = \tilde{\gamma}\tilde{U}_x$,

$$\begin{aligned} \frac{1}{f} \frac{\partial \mathcal{H}}{\partial T} &= -(q^d + T\partial_T q^d) + (1 - T\partial_x q^d - \tilde{\chi})q^d + \tilde{\Lambda} l \partial_l q^d + \tilde{\chi} q^d \\ &= -T (\partial_T q^d + q^d \partial_x q^d) + \tilde{\Lambda} l \partial_l q^d = -T s_{qq} + \tilde{\Lambda} l \partial_l q^d, \end{aligned}$$

where the q^d terms cancel in pairs, the incentive terms $\tilde{\chi}q^d$ cancel exactly, and the Slutsky identity recombines the Marshallian price and income responses into the compensated s_{qq} . The cancellation of the incentive terms is the mechanism-design counterpart of a familiar fact: the utility cost of the price to inframarginal purchases (Roy) is exactly offset by its incentive value, because the mechanism re-optimizes compensation, and only the substitution distortion Ts_{qq} against the screening benefit $\tilde{\Lambda}l\partial_l q^d$ survives.

Aggregating to date 0 (the R^{-t} and transition-density weights convert the pointwise integral into the population expectation $\mathbb{E}_0$, as in the social-cost-of-carbon derivation of appendix B.1) and setting the derivative to zero,

$$T_t \mathbb{E}_0[s_{qq,t}] = \mathbb{E}_0 [\tilde{\Lambda}_t l_t \partial_l q_t^d].$$

Substitute $l \partial_l q^d = \rho U_c(\gamma_q - \gamma_c)/|\mathcal{D}| = \rho s_{qq}(\gamma_c - \gamma_q)$, using $s_{qq} = -U_c/|\mathcal{D}|$,

$$T_t \mathbb{E}_0[s_{qq,t}] = \rho \mathbb{E}_0 [s_{qq,t}(\gamma_{c,t} - \gamma_{q,t})\tilde{\Lambda}_t] \quad \implies \quad \frac{T_t}{\rho} = \frac{\mathbb{E}_0[|s_{qq,t}|(\gamma_{c,t} - \gamma_{q,t})\tilde{\Lambda}_t]}{\mathbb{E}_0[|s_{qq,t}|]},$$

dividing by $\rho \mathbb{E}_0[s_{qq}]$ (both sides negative). Since $\rho = 1 + T = 1 + \tau^a$, the left-hand side is $\tau^a/(1 + \tau^a)$, which is Result 6.3.2. At node-specific prices the same steps without the integral give $T(\theta^t)/\rho = (\gamma_c - \gamma_q)\tilde{\Lambda}$, which is the unconstrained wedge of appendix A.3; the uniform tax is its $|s_{qq}|$ -weighted average. Under weak separability every node wedge is zero, so $T_t = 0$ (Corollary 6.3.1); in retirement $\psi \equiv 0$ so $\tilde{\Lambda} \equiv 0$ and $T_t = 0$ (Corollary 6.3.2).

C.6 Savings wedge (Result 6.3.3)

From the FOC for x , using $T\partial_x q^d = (T/\rho)(\rho\partial_x q^d) = \varpi m_q$,

$$\frac{\phi}{f} = \frac{1 - T\partial_x q^d - \tilde{\gamma}\tilde{\Lambda}}{\tilde{U}_x} = \frac{1 - \chi^F}{\tilde{U}_x}, \quad \chi^F = \tilde{\gamma}\tilde{\Lambda} + \varpi m_q.$$

The FOC for w gives $\lambda_{1,t+1} = \beta R \phi / f = \beta R (1 - \chi_t^F) / \tilde{U}_{x,t}$. Integrating the costate equation over $(0, \infty)$ with $\psi(0) = \psi(\infty) = 0$, and using $\int \bar{\alpha}_t f d\theta = 1$ and $\int f_2 d\theta = 0$,

$$\lambda_{1,t} = \int_0^\infty \phi d\theta = \mathbb{E}_t \left[\frac{1 - \chi_t^F}{\tilde{U}_{x,t}} \right].$$

Evaluating at $t + 1$ and equating with the FOC for w ,

$$\frac{1 - \chi_t^F}{\tilde{U}_{x,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}^F}{\tilde{U}_{x,t+1}} \right] \iff \frac{1}{\widehat{\tilde{U}}_{x,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{\tilde{U}}_{x,t+1}} \right],$$

with $\widehat{\tilde{U}}_x = \tilde{U}_x / (1 - \chi^F)$. The exact wedge, its second-order approximation, and the lifetime formula follow the steps of appendix A.2 with $\chi_t \mapsto \chi_t^F$; in particular $\tau_t^s \approx \text{Var}_t(\ln \tilde{U}_{x,t+1}) + (\chi_t^F - \mathbb{E}_t \chi_{t+1}^F) + \text{Cov}_t(\chi_{t+1}^F, \ln \tilde{U}_{x,t+1})$.

C.7 Comprehensive-wedge decomposition (Result 6.3.4)

Substituting $\phi = (1 - T\partial_x q^d)f / \tilde{U}_x - \psi\tilde{\gamma}/\theta$ into the costate equation gives the linear ODE

$$\dot{\psi} - \frac{\tilde{\gamma}}{\theta}\psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{(1 - T\partial_x q^d)f}{\tilde{U}_x},$$

which is the ODE of appendix A.1 with the forcing term f/U_c replaced by $(1 - T\partial_x q^d)f / \tilde{U}_x$. Solving with the integrating factor $\exp[-\int \tilde{\gamma}(\tilde{x})d\tilde{x}/\tilde{x}]$ and applying the identity of appendix A.1, now with a single composite consumption object,

$$\frac{\tilde{U}_x(\theta)}{\tilde{U}_x(x)} \exp \left[-\int_\theta^x \tilde{\gamma}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] = \exp \left[\int_\theta^x \left(\bar{\sigma}(\tilde{x}) \frac{\dot{x}(\tilde{x})}{x(\tilde{x})} - \tilde{\gamma}(\tilde{x}) \frac{\dot{y}(\tilde{x})}{y(\tilde{x})} \right) d\tilde{x} \right]$$

(the cross-good term $\sigma_{cq}\dot{q}/q$ of appendix A.1 is absent because the reduced problem assigns a single expenditure object). The static labor condition of appendix C.4 gives $\psi_{t-1}/f_{t-1} = \theta_{t-1} \Omega_{t-1} / (\tilde{A}_{t-1} \tilde{U}_{x,t-1})$, so the $\lambda_{2,t} = \beta R \psi_{t-1} / f_{t-1}$ block produces the persistence term $\beta R \Omega_{t-1} \tilde{D}_t$, in which the recursing object is the comprehensive wedge Ω . Collecting terms,

$$\Omega_t = \tilde{A}_t(\theta)\tilde{B}_t(\theta)\tilde{C}_t(\theta) + \beta R \Omega_{t-1}\tilde{D}_t(\theta),$$

$$\tilde{A}_t(\theta) = 1 + \tilde{\epsilon}_t(\theta) - \tilde{\gamma}_t(\theta)$$

$$\tilde{B}_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$\tilde{C}_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\tilde{\sigma}_t(\tilde{x}) \frac{\dot{x}_t(\tilde{x})}{x_t(\tilde{x})} - \tilde{\gamma}_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] \left(1 - T_t \partial_x q^d(x) - \lambda_{1,t} \bar{\alpha}_t(x) \tilde{U}_{x,t}(x) \right) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$\tilde{D}_t(\theta) = \frac{\tilde{A}_t(\theta)}{\tilde{A}_{t-1}} \frac{\tilde{U}_{x,t}(\theta)}{\tilde{U}_{x,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \tilde{\gamma}_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}.$$

The revenue term $-T_t \partial_x q^d$ inside the $\tilde{C}_t$ weight is the only trace of the second good in the redistributive valuation; it reflects that extracting a unit of expenditure from a higher type also moves commodity revenue.

C.8 Welfare cost of uniformity (Result 6.4.1)

At node-specific prices the derivative computed in appendix C.5, before aggregation, is the per-node resource residual (in date- t resources, per unit density)

$$\left. \frac{1}{f} \frac{\partial \mathcal{H}}{\partial T} \right|_{\text{node}} = -T s_{qq} + \tilde{\Lambda} l \partial_l q^d = \rho s_{qq} (\varpi^* - \varpi) = \rho |s_{qq}| (\varpi - \varpi^*),$$

writing $\varpi = T/\rho$ for the node's ad-valorem wedge and $\varpi^* = (\gamma_c - \gamma_q) \tilde{\Lambda}$ for its optimal value (the residual vanishes at $\varpi = \varpi^*$, reproducing appendix C.5). The marginal resource cost of moving the node's wedge is $\partial \mathcal{H} / \partial \varpi = \rho \partial \mathcal{H} / \partial T$ (since $dT = \rho d\varpi$ to leading order), so the loss from setting ϖ rather than ϖ^* at a node is, to second order (the envelope theorem killing the first-order effects of the induced adjustments in (x, y) and the multipliers),

$$\int_{\varpi^*}^{\varpi} \rho^2 |s_{qq}| (v - \varpi^*) dv = \frac{1}{2} \rho^2 |s_{qq}| (\varpi - \varpi^*)^2.$$

Aggregating over histories and minimizing the resulting quadratic over the common ϖ yields the weighted mean $\varpi = \mathbb{E}_t^{|s|}[\varpi_t^*]$ of Result 6.3.2 and, at that minimizer,

$$\text{Loss}_t \approx \frac{1}{2} \rho^2 \mathbb{E}_0[|s_{qq,t}|] \text{Var}_t^{|s|}(\varpi_t^*), \quad \rho = 1 + T_t,$$

where $\mathbb{E}_t^{|s|}$ and $\text{Var}_t^{|s|}$ are the $|s_{qq,t}|$ -weighted population mean and variance over date- t histories.

C.9 The externality

With the externality of appendix B.1, the stock is aggregate and deterministic, hence invariant to reports, and acts as a dated preference shifter through $\tilde{U}(x, l; T, S_t)$. Pricing emissions at $\zeta_t = \kappa R^{-t} \text{SCC}_t$ converts the resource cost of the dirty good to the emissions-inclusive producer price $1 + \text{SCC}_t$, so the resource cost is $c + (1 + \text{SCC}_t)q = x - (\tau_t^q - \text{SCC}_t)q^d = x - T_t q^d$ with the screening tax $T_t = \tau_t^q - \text{SCC}_t$. Every FOC in (x, l, w, w_2) is then that of appendix C.3 with T_t the screening component and all statistics evaluated at (x, l, T_t, S_t) , so Results 6.3.1–6.4.1 hold verbatim and the optimal uniform price splits additively,

$$1 + \tau_t^q = 1 + \text{SCC}_t + T_t^{\text{screen}}, \quad \frac{T_t^{\text{screen}}}{1 + \tau_t^q} = \frac{\mathbb{E}_0[|s_{qq,t}|(\gamma_{c,t} - \gamma_{q,t})\tilde{\Lambda}_t]}{\mathbb{E}_0[|s_{qq,t}|]}.$$

The social cost of carbon is unchanged in form from appendix B.1, with $U_c \mapsto \tilde{U}_x$, the fiscal factor built from the composite statistics $(\tilde{\epsilon}, \tilde{\gamma})$, and $\hat{\tilde{U}}_x$ in place of U_c in the compensation term. The corrective component is uniform across types and histories because damages depend only on aggregate emissions, so the emissions margin carries no information about types and cannot screen; the uniformity restriction therefore binds only the Corlett–Hague screening component and never the Pigouvian one.